

Module 2: Diagnostic/ Clinical Microbiology

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DIAGNOSTIC/CLINICAL MICROBIOLOGY

• a field of microbiology
concerned with the etiologic
diagnosis of infection

etiologic = what causes the disease and the manner of causation

Major Objectives of the Microbiologist In this Laboratory :

1. To **isolate** and **identify** the microorganisms causing the disease. (know the disease)
2. To determine the **susceptibility** of these pathogens to **antimicrobial drugs** for proper eradication or for treatment of the disease.

(iradicate the pathogen)

Close communication between the:

PHYSICIAN

Ensures proper sample is collected

Gives guidance by way of a clinical diagnosis

MICROBIOLOGIST

Proper collection and transport; diagnostic methods

Feedback for proper treatment



Universal Precautions Inside a Diagnostic Laboratory

1. Use of appropriate **barrier precautions** to prevent skin and mucous membrane exposure. This is to avoid sample contamination
2. Thorough **washing of hands** and other skin surfaces after gloves are removed and immediately after any contamination.
3. Taking special care to **avoid injuries with sharp objects** such as needles and scalpels



Personal Protective Equipment (PPE)

Proper Handwashing Technique

<https://www.youtube.com/watch?v=oGQpJafaWj0>

4. Use of mouthpieces or other ventilation devices with 1 way-valves for performing any resuscitation measures. This is to avoid mouth-to-mouth

5. Refraining from handling patient care equipment or patients if health care workers have exudative lesions or weeping dermatitis.

Exudative lesions are pus-like fluid from circulatory system. Weeping dermatitis are pus-like fluid from blisters caused by Staphylococcus aureus

6. Very stringent adherence to precautions by pregnant workers. This is to avoid nosocomial diseases to protect the baby.

Safety Precautions For Lab. Workers

1. Placement of **specimens in leak-proof, sealed containers** for transport and avoidance of contamination of specimen container surfaces.
2. **Changing of gloves and washing of hands after specimen processing.**

Before and after handling biological samples

3. Limiting the use of needles and syringes to those situations in which there are no alternatives.

4. Thorough decontamination* of any equipment or material that has been contaminated with any blood or body fluid before it is repaired or transported.

*** DECONTAMINATION:**

→ if applicable : ~ by **steam sterilization
normally at 15 psi (121°C) for 30-45 min
or even up to 1 hr for samples with
potentially greater hazard**

* DECONTAMINATION:

- kill all possible life forms

→ for workbenches/surfaces : washing with a liquid disinfectant:

- liquid disinfectants does not kill all life forms as it can leave those that are spore-forming

phenolic compounds

- mutagenic (can cause cancer)

70% alcohol, QUATS

0.5% Na hypochlorite

10% household bleach, etc.

General Guidelines on Specimen Collection and Processing (Mandell et al., 1995)

1. The specimen selected should be **representative** of the disease process.

For instance, testing for TB should have sputum sample

2. An **adequate quantity** of the material should be obtained for complete examination.

3. **Scrupulous attention must be given to **avoiding contamination** of the specimen by the many varieties of organisms indigenous to the skin and mucous membranes, culture of which may often be more confusing and misleading than helpful.**

4. **The material should be **forwarded promptly** to the laboratory (may use transport media)**

Especially for fastidious organisms which have a special ingredient in their diet and cannot grow on general media

5. Specimens should be obtained before antimicrobial agents have been administered, and in the case of viral infections, early in the acute phase of the illness to enhance recovery of the virus.

This is to avoid possibility of low yield of microorganisms.

- Acute phase: start of disease
- Convalescent phase: recovery

Specimen Collection & Transport

- Use of a sterile container :



- disposable, sterile plastic cups
- swab (cotton-, alginate- or polyester-tipped)
- commercially available systems (tube with swab, ampoule or pouch with holding medium, etc.)

- commercially available in groceries for cleaning have antimicrobials



- **Availability of a Variety of Transport Media :**

Stuart's Medium (pH is maintained; no nutrients, prevents dehydration of secretions and enzymatic self-destruction of pathogens)

"self-destruction" - means that it dies on its own toxins



Cary-Blair Transport Medium (~ for fecal specimens; for recovery of *Salmonella*, *Shigella*, *Vibrio*, *Campylobacter* etc.)

- vibrio is not naturally found in the body



Amies Charcoal Medium (for the recovery of *N. gonorrhoeae* & other fastidious organisms)

- charcoal neutralizes toxins to prevent death



How to use a swab-transport medium set-up

(<https://www.youtube.com/watch?v=vOjAWgSJvh4>)

- **Processing of the specimen as soon as possible :**

- especially for :
 - CSF samples ; should not be frozen; CSF samples are collected for those suspected with meningitis
 - specimens with gonococci and *Bordetella pertussis*

- *Bordetella pertussis* causes whooping cough, which is uncontrollable coughing until oxygen is depleted

- if not possible : refrigeration at 4 – 6°C
(for many clinical samples)
: immediate freezing (not applicable to CSF)

General Techniques/Procedures

- **Microscopy** - least sensitive technique; we cannot identify microorganism based on morphology alone
- Culture isolation and pathogen ID
- Detection of antigen(s)/antibodies using immunologic assays
- Detection of genes specific for the pathogen in specimens obtained from the patient (molecular method)
- Other Tests

Step 1

Result:
All cells purple

Flood the heat-fixed smear with crystal violet for 1 min

Step 2

Result:
All cells remain purple

Add iodine solution for 1 min

Step 3

Result:
Gram-positive cells are purple;
gram-negative cells are colorless

Decolorize with alcohol briefly — about 20 sec

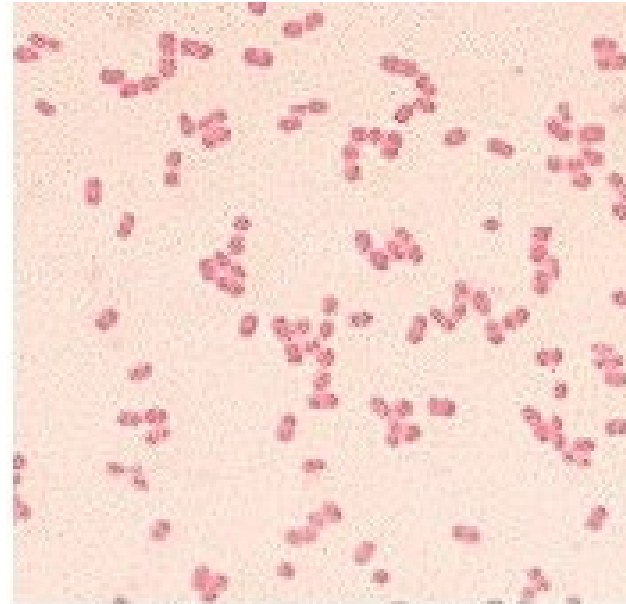
Step 4

Result:
Gram-positive (G^+) cells are purple;
gram-negative (G^-) cells are pink to red

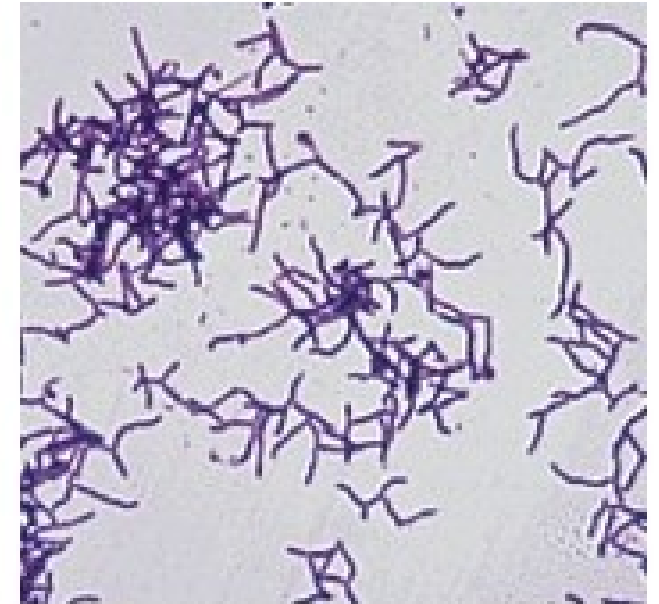
Counterstain with safranin for 1–2 min

Gram staining

- differential staining



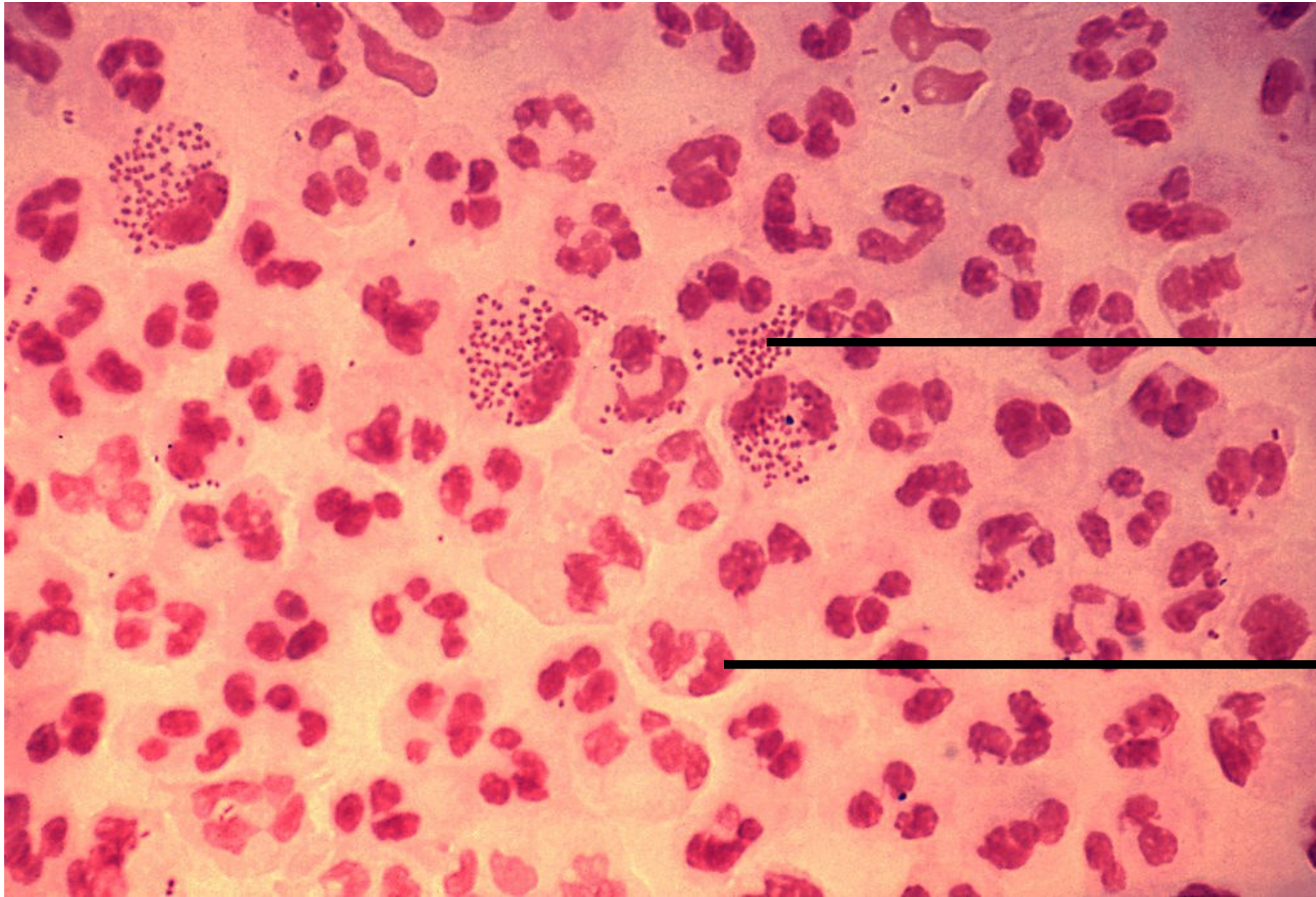
Gram Negative



Gram Positive

https://maxbioblog.weebly.com/uploads/9/1/5/3/91531616/20150515114749001_orig.jpg

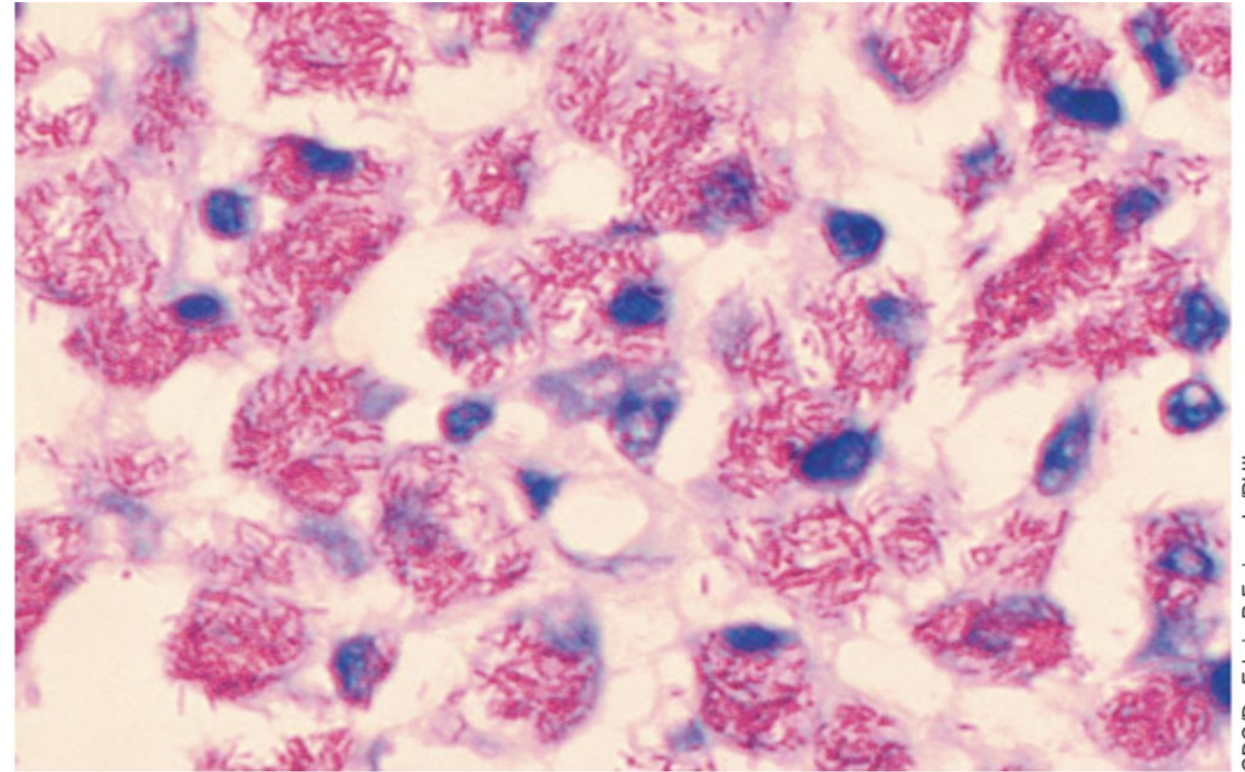
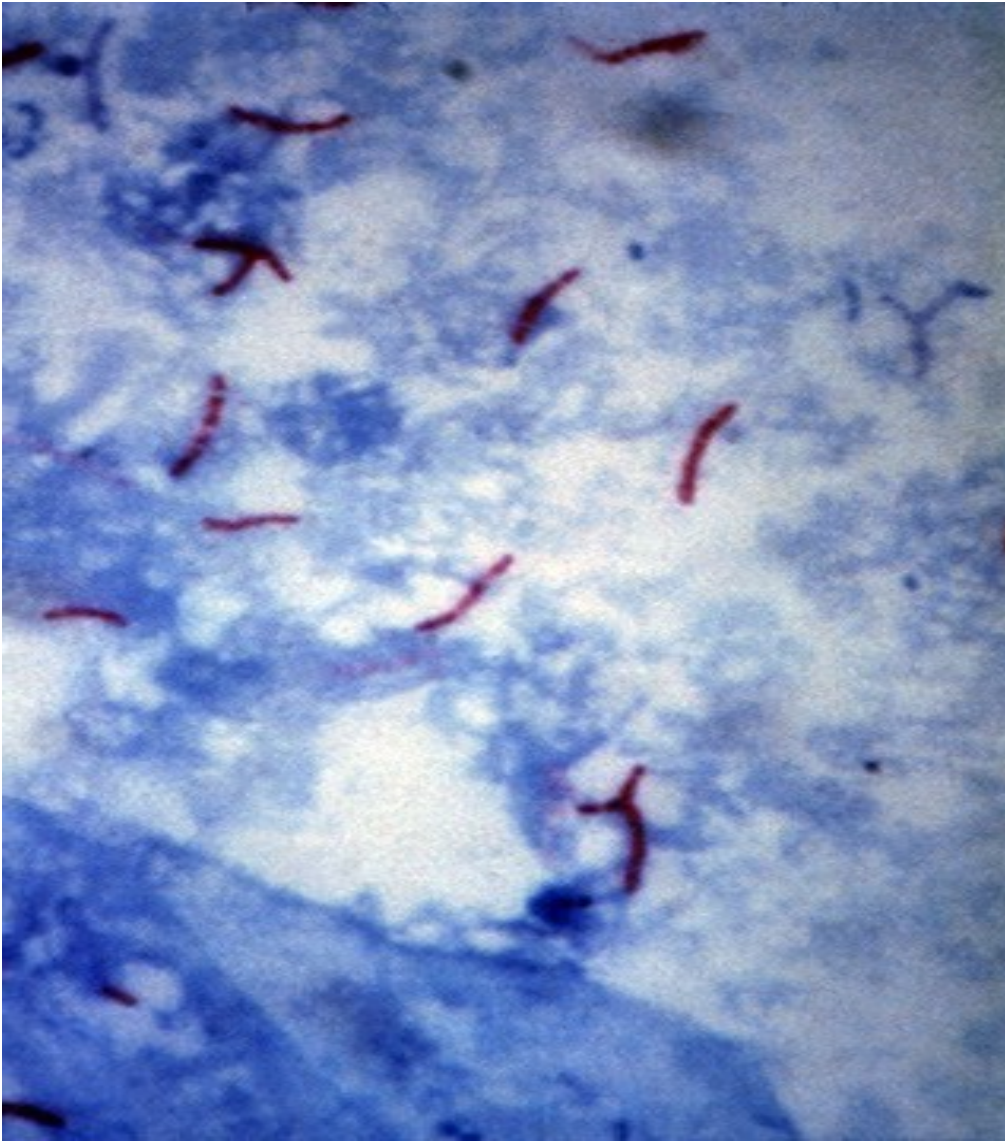
Gram positive have thicker peptidoglycan



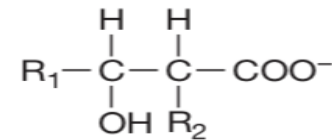
N. gonorrhoeae

Leukocyte
nuclei

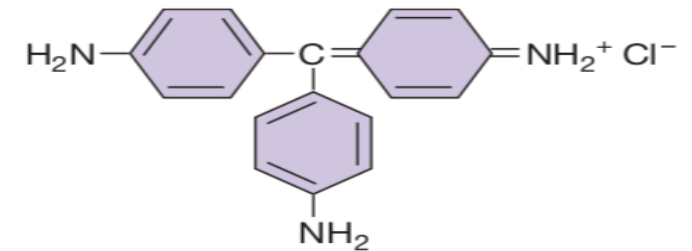
Gonorrhea smear made from a pus sample



CDC/Dr. Edwin P. Ewing, Jr./PHIL



(a) Mycolic acid; R_1 and R_2 are long-chain aliphatic hydrocarbons



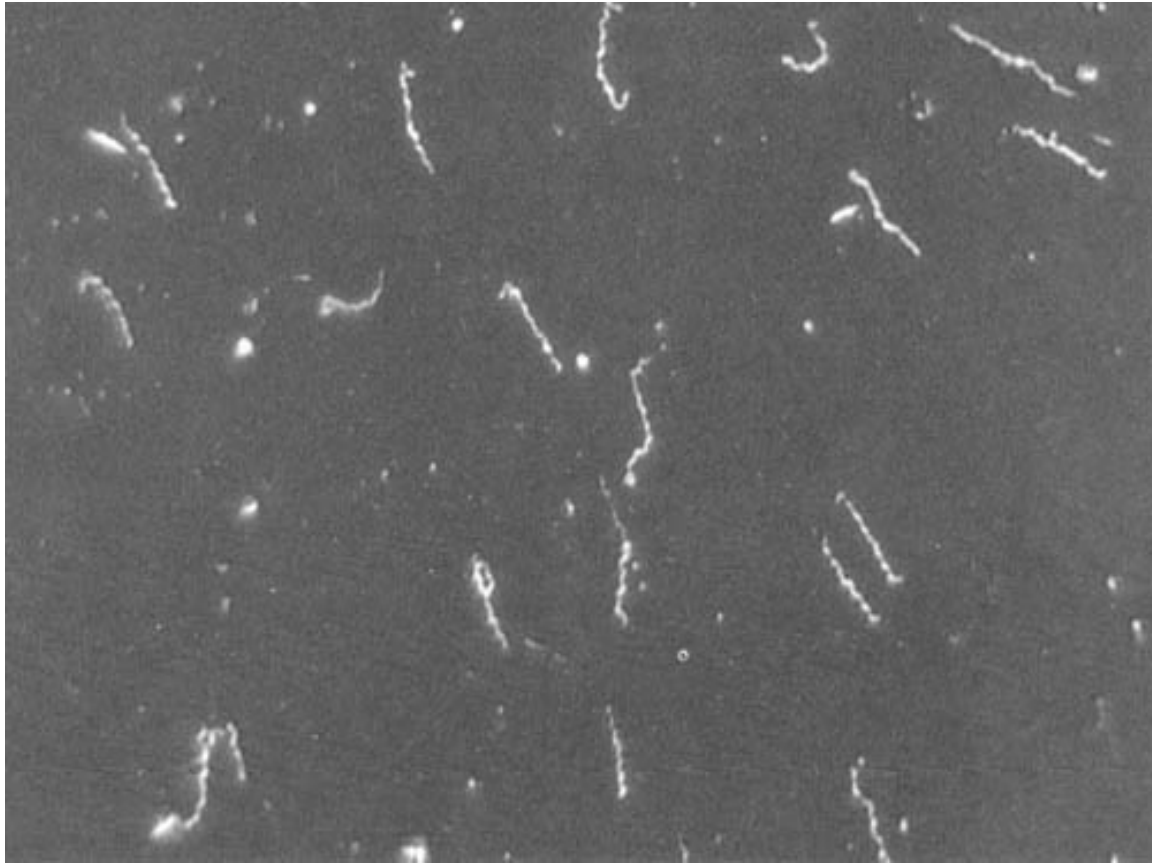
(b) Basic fuchsin

Acid Fast Stain (Ziehl-Neelsen stain) of *M. tuberculosis* (red)



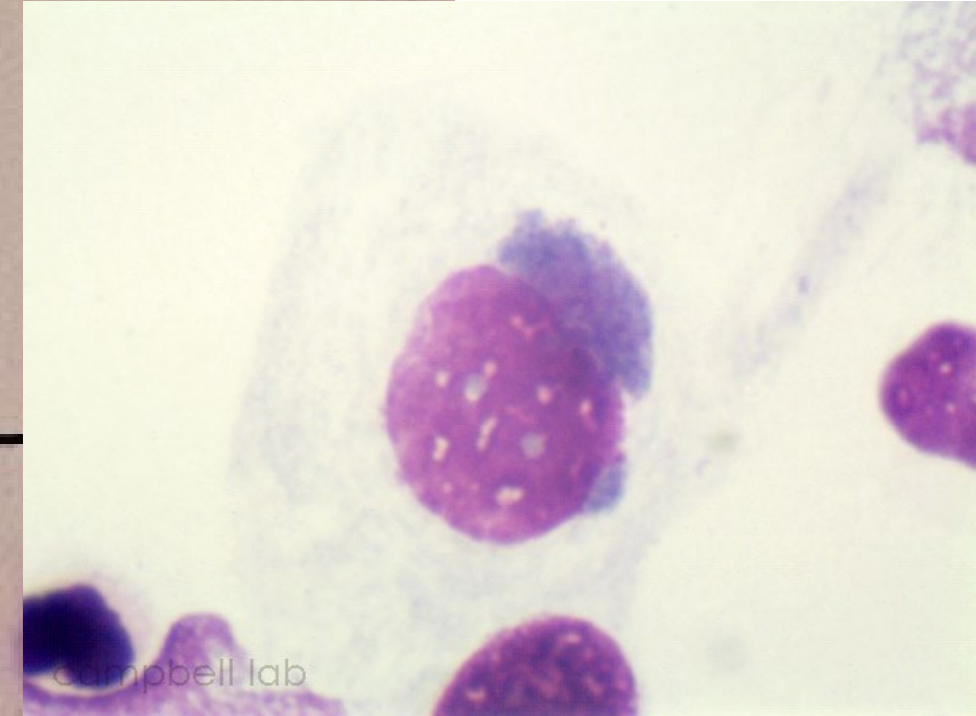
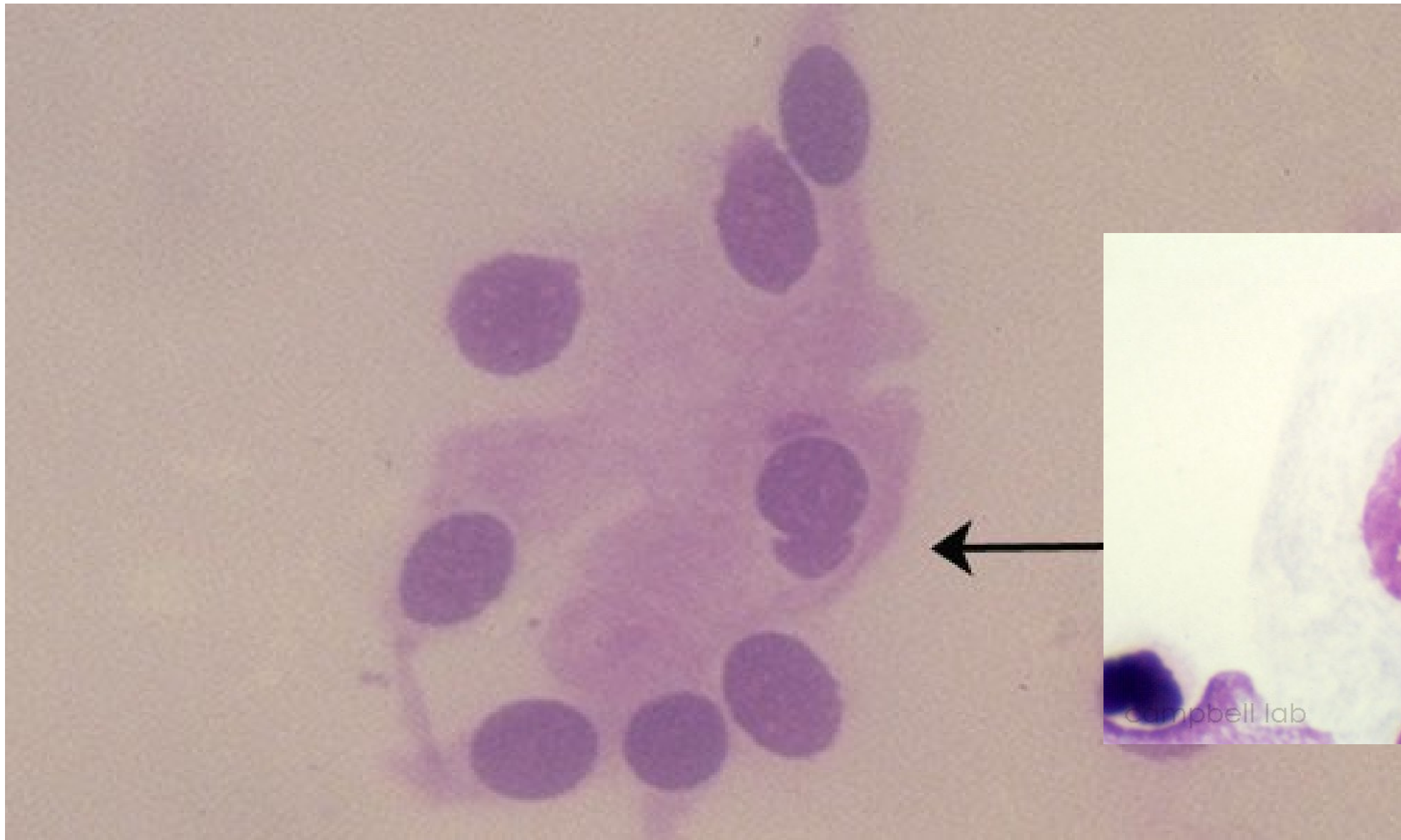
Acid-Fast Staining

(<https://www.youtube.com/watch?v=A4IZLYVXGaA>)



Darkfield Microscopy (*Treponema sp.*)

- dark field of vision
- light enters from sides
- sample is not stained



<https://www.google.com.ph/search?q=giemsa+stain+chlamydia>

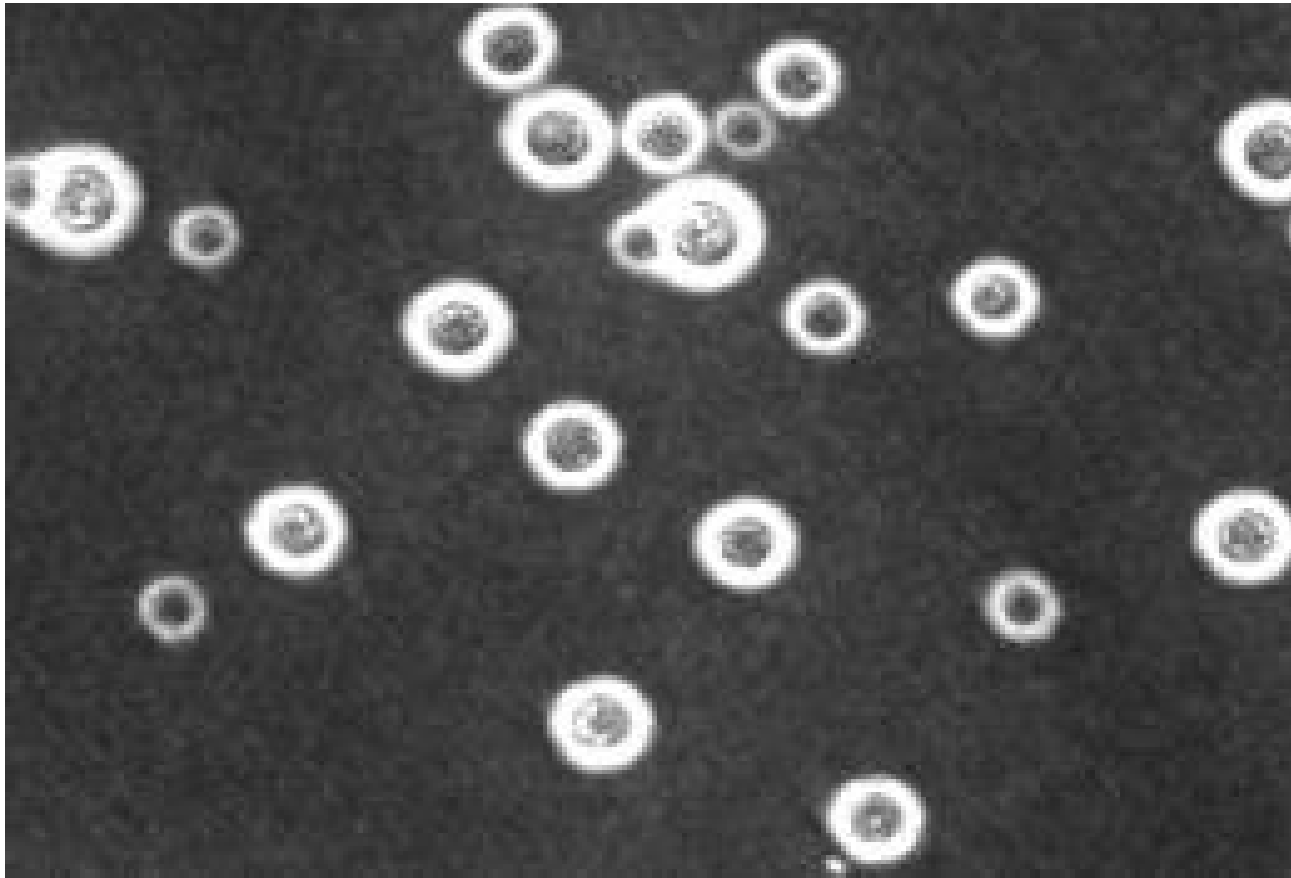
Giemsa stain is differential staining

Giemsa stain of *Chlamydia* (EB) : conjunctival scraping

KOH Preparation

Richard P. Usatine, MD

The Journal of Family Practice



Negative Stain : *Cryptococcus neoformans*
(e.g. India ink)

samples are naturally negatively charged too to repel the stain



Negative Stain : *Saccharomyces cerevisiae*
(e.g. Nigrosin)

Select a fresh blister on the patient.

- Using a blunt scalpel blade, gently deroof the lesion
- Scrape the base of the lesion
- Smear the tissue onto a clean microscope slide
- Allow it to dry in the air
- Fix the specimen with preservative

Taking a Tzanck smear



Deroof blister



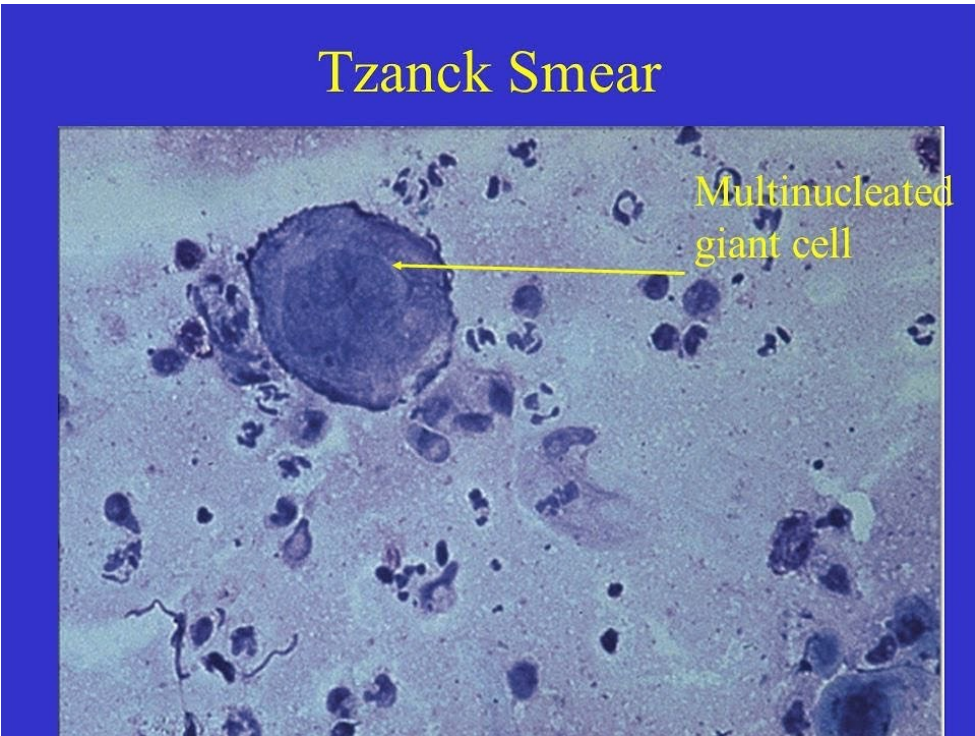
Smear onto slide



Air dry



Add fixative



Tzanck smear of a herpesvirus infection: enlarged, multinucleated giant cells

- this stain uses giemsa stain and methylene blue
- for chicken pox cells

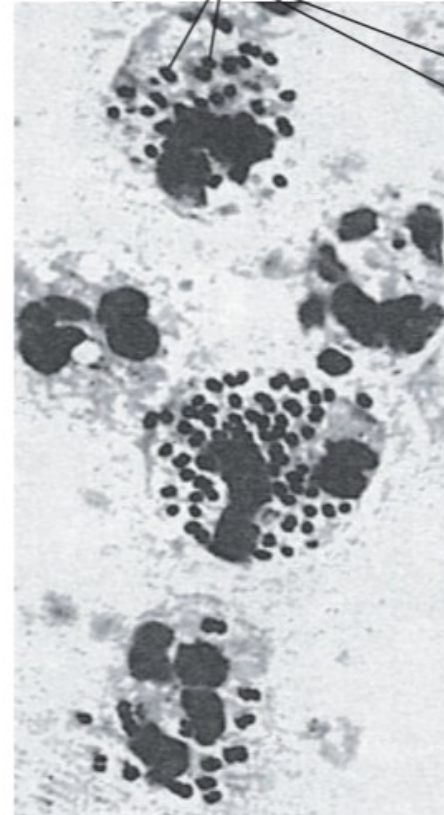
General Techniques/Procedures

- Microscopy
- Culture isolation and pathogen ID
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- Other Tests

Growth-dependent Diagnostic Methods

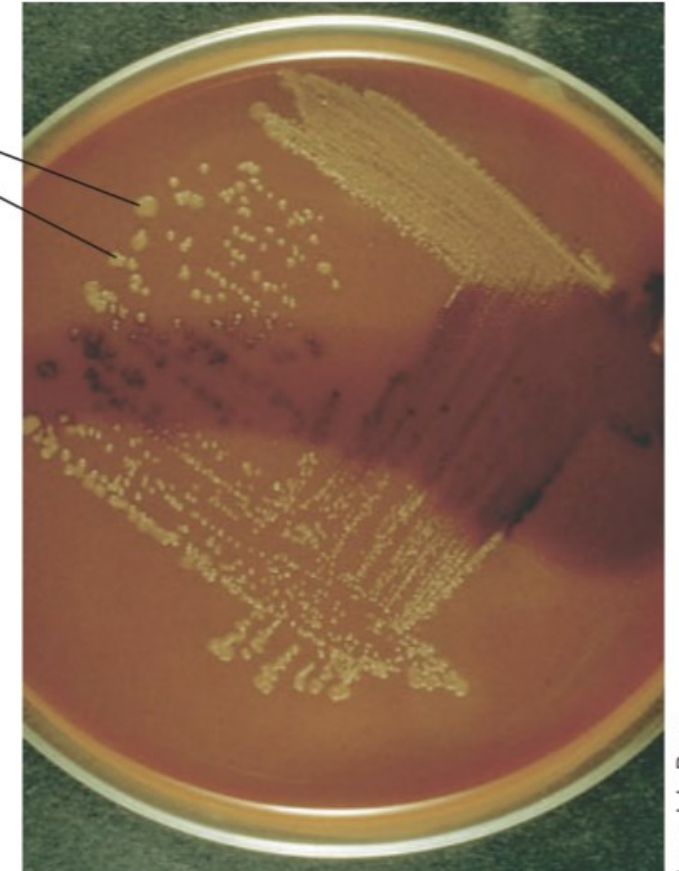


Cells and colonies of
N. gonorrhoeae



Theodor Rosebury

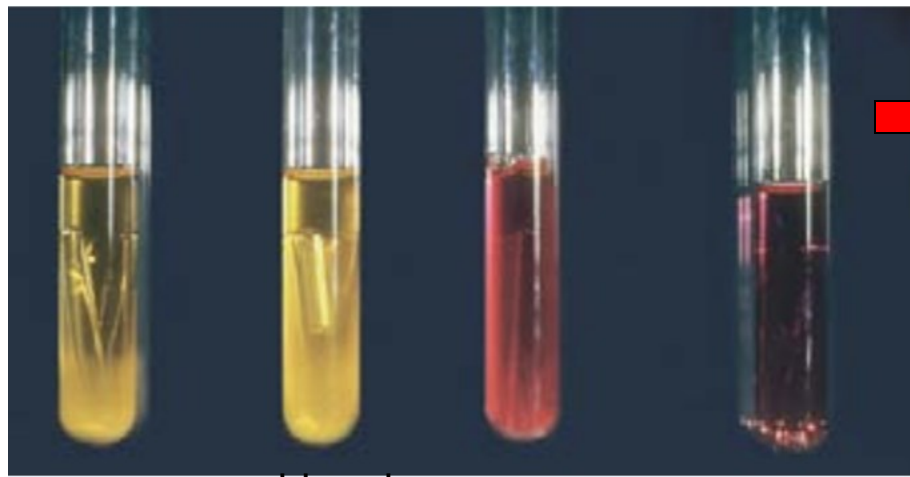
(a)



Leon J. LeBeau

(b)

Culture Isolation and Identification



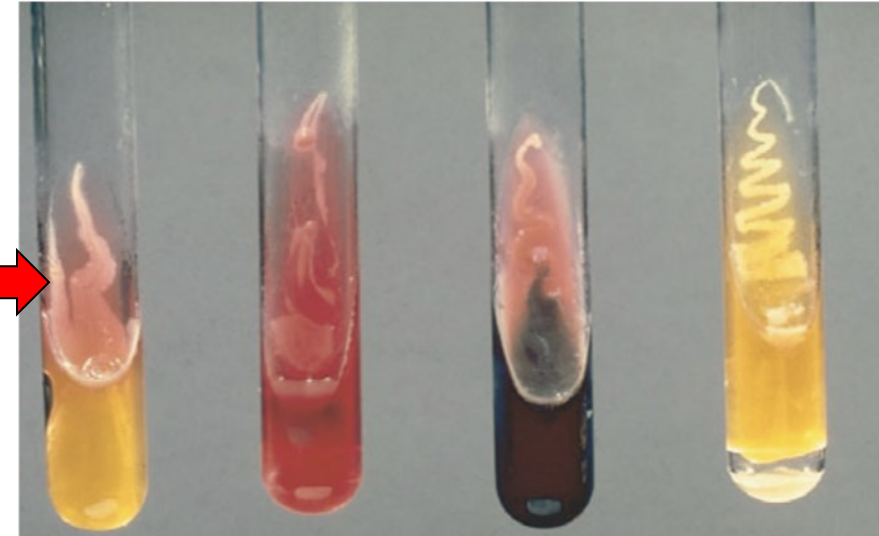
acid acid and gas (-) uninoculated

metabolism - utiliza

Use of a differential medium to assess sugar fermentation

- color indicator
- differential medium: metabolism of sugar
- negative control is a must and uninoculated

A conventional diagnostic test for enteric bacteria in triple sugar iron (TSI) agar

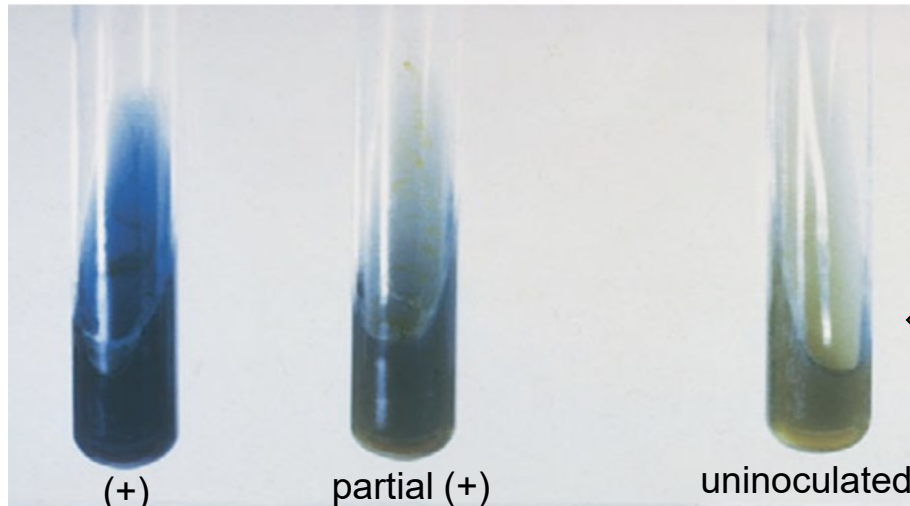


Ferm. Glu. only No rxn H₂S form'n Ferm. Glu. + another sugar

- reagent: phenol red w/ ferrous ion

- with crack in the butt = gas formation, pushing media upward

- yellow with red above = incomplete fermentation



(+) partial (+) uninoculated

Measurement of citrate utilization by *Salmonella* on Simmons citrate agar

- reagent: bromothymol blue
- detects shift in pH towards alkalinity

Growth-dependent diagnostic methods used for the identification of clinical isolates by color changes in various diagnostic media.



API = ANALYTICAL PROFILE INDEX

- tests what type of sugar is metabolized
- faster turnaround time
- one tube = test per sugar (before)
- multiple miniature containers (4 microorganisms per test)



Media kits used for the rapid identification of clinical isolates

Another arrangement of a miniaturized test kit

Growth-dependent diagnostic methods used for the identification of clinical isolates by color changes in various diagnostic media.

CLINICAL SPECIMENS (For Culture)

- **Blood**
- **Urine**
- **Fecal Samples**
- **Wounds, Abscesses and Pus Samples**
- **Cerebrospinal Fluid**
- **Respiratory Secretions**
- **Genital Specimens**

1. Blood

- common in hospitals, easy to know multiple tests in a human (drugs, diseases, etc.), easy overview of one's health

❑ **its culture : one of the most important procedures in clinical microbiology**



1. Blood

- ❑ **success of blood culture depends on :**
 - method of blood collection**
 - timing of blood collection ***
 - volume of blood collected**, etc.**

*** re: timing of blood collection**

□ e.g. chronic vs. intermittent septicemia

septicemia = “widespread destruction
of tissues due to
absorption of disease-
causing bacteria or
their toxins from the
bloodstream”
= a form of blood poisoning

- chronic septicemia: = blood infection persists in a long period of time
- intermittent septicemia = occurs in irregular intervals, not all the time has spiking

**** re: volume of blood collected**

☐ for blood, a large volume should be collected for culture (~10 – 20 ml)

**Why? > 50% of all septic patients have
< 1 organism/ml**



1. Blood

Standard blood collection procedure:

Skin disinfection



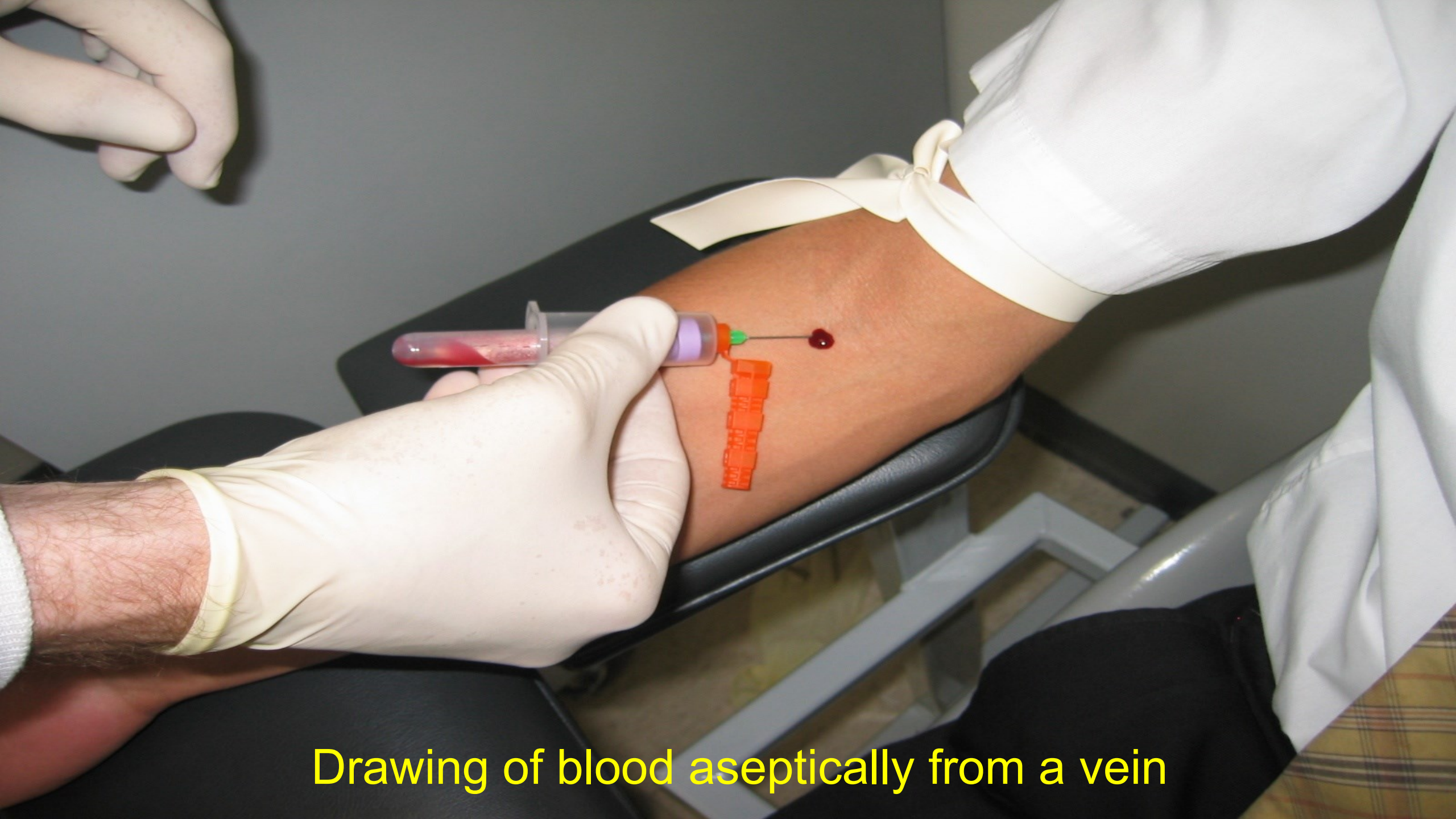
Remove blood aseptically from a vein



Inject into a blood culture bottle with a rich, all purpose culture medium (e.g. Blood Agar)



35-37°C , ~ 7 days



Drawing of blood aseptically from a vein



Two blood culture bottles containing an anticoagulant and a general-purpose culture medium

This is agar is used to know if microorganisms is capable of breaking down hemoglobin
Use of human blood is unethical, so small animals like sheep and rabbit are used (added small % in agar)



ASM MicrobeLibrary.org © Buxton

Blood Agar

1. Blood

❑ **microorganisms usually detected by:**

Microscopy

subculture

turbidimetric methods

molecular methods

serological methods, etc.

2. Urine

❑ one of the most frequently submitted specimens for culture (~ 5 ml)

- collect midstream urine
- initial urine have NMF which can result to false positive



2. Urine

☐ the 1st portion of urine voided must be discarded; midstream urine; “clean catch specimen”

reason : microbial colonization of the urethra

2. Urine

☐ **delays in transport of the urine should be avoided**

**reason : urinary tract pathogens can also grow
in urine**

**otherwise : refrigerate and place in a
bacteriostatic urine preservative**

2. Urine

☐ fairly high cell numbers are found in UTI since urine can support extensive bacterial growth

☐ for “significant” bacteriuria :

$\geq 10^5$ organisms/ml or

≥ 2 bacteria/OIO field

$\geq 10^4$ bacteria/ml on solid media

2. Urine

☐ if simply contaminated from the external genitalia :

$< 10^3$ organisms/ml

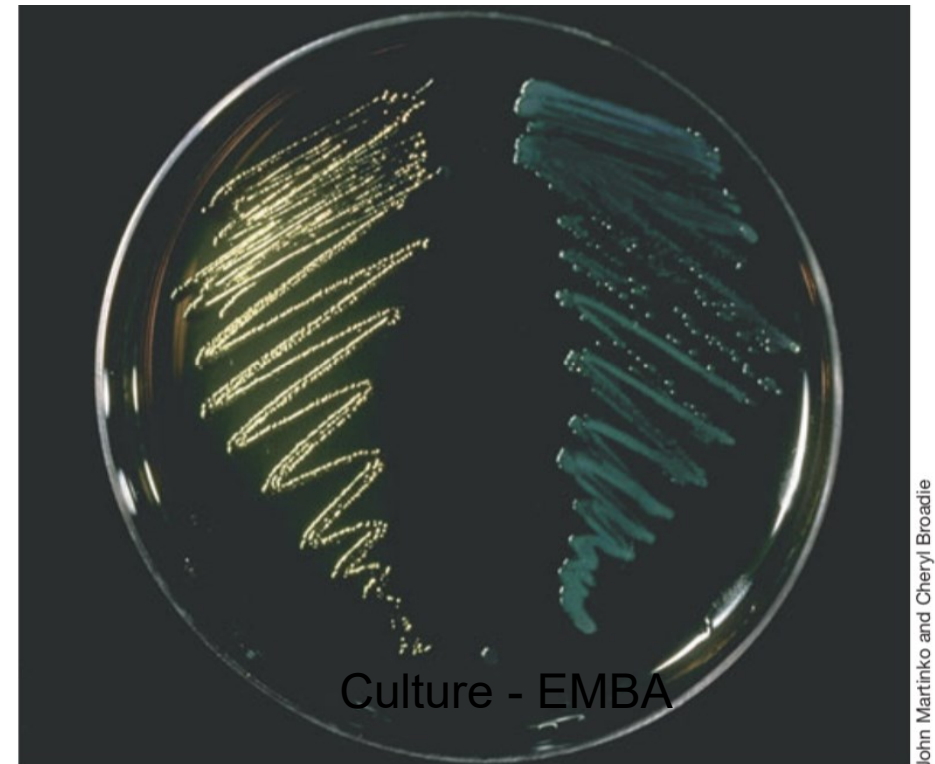
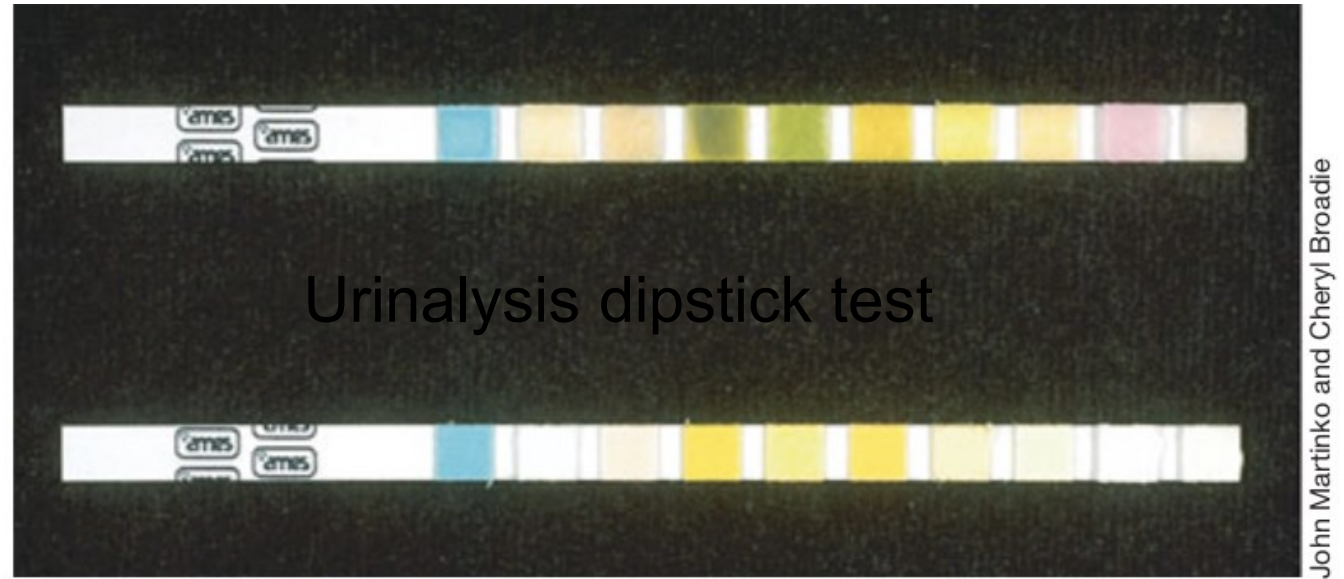
2. Urine

- ❑ most common UT pathogen : *E. coli*
- ❑ other pathogens :
 - Klebsiella*
 - Enterobacter*
 - Proteus*
 - Pseudomonas*
 - Staphylococcus*
 - Enterococcus faecalis*, etc.

2. Urine

□ microorganisms usually detected by:

- Microscopy
- Urinalysis dipstick test
- Culture
- Gram staining



3. Fecal/Stool Samples

- ❑ collected for the detection/culture of some enteric pathogens

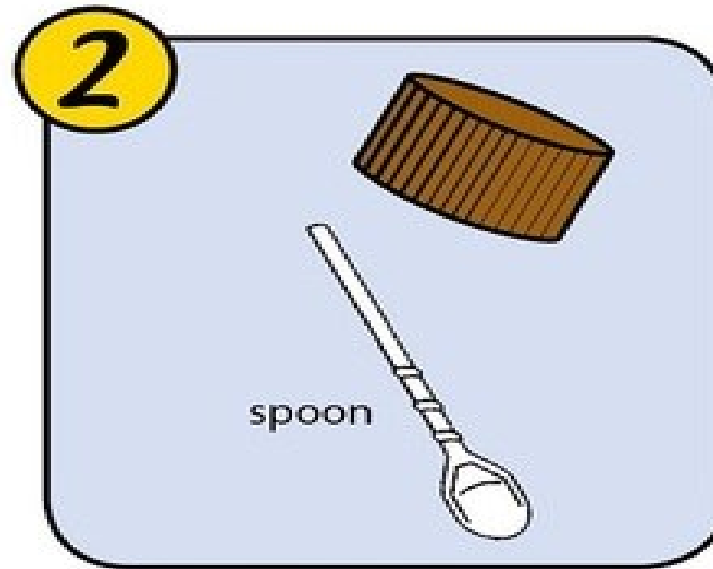


3. Fecal/Stool Samples

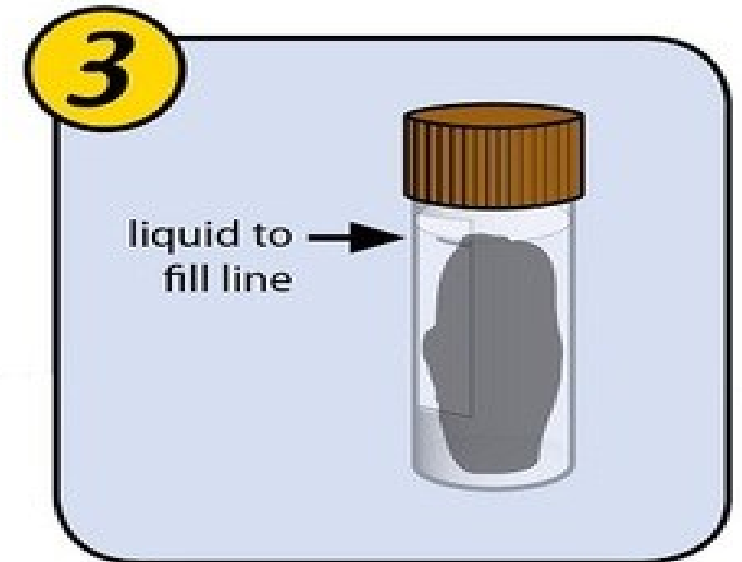
- ❑ should be properly collected and preserved (mixed with a preservative like phosphate buffer with glycerol in a transport medium)



Collect on plastic wrap and transfer to vial until liquid reaches fill line.



Remove spoon from lid and discard.



Replace cap on vial tightly and shake for a minute. Place vial in refrigerator until ready to ship.

3. Fecal Samples

- ❑ delays in transport to the laboratory should be avoided to prevent acidic changes in the stool (a result of bacterial metabolism) → toxic to:

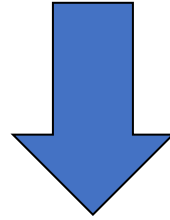
Shigella

Salmonella

etc.

4. Fecal Samples

❑ when foodborne or waterborne infections are suspected



several selective media should be inoculated for the isolation of specific bacteria → **rectal swab sample is not recommended**

Table 31.2 Colony characteristics of frequently isolated gram-negative rods cultured on various clinically useful media

Organism	Agar media ^a				
	EMB	MC	SS	BS	HE
<i>Escherichia coli</i>	Dark center with greenish metallic sheen	Red or pink	Red to pink	Mostly inhibited	Yellow-pink
<i>Enterobacter</i>	Similar to <i>E. coli</i> , but colonies are larger	Red or pink	White or beige	Mucoid colonies with silver sheen	Yellow-pink
<i>Klebsiella</i>	Large, mucoid, brownish	Pink	Red to pink	Mostly inhibited	Yellow-pink
<i>Proteus</i>	Translucent, colorless	Transparent, colorless	Black center, clear periphery	Green	Clear
<i>Pseudomonas</i>	Translucent, colorless to gold	Transparent, colorless	Mostly inhibited	No growth	Clear
<i>Salmonella</i>	Translucent, colorless to gold	Translucent, colorless	Opaque	Black to dark green	Green or transparent with black centers
<i>Shigella</i>	Translucent, colorless to gold	Transparent, colorless	Opaque	Brown or inhibited	Green or transparent

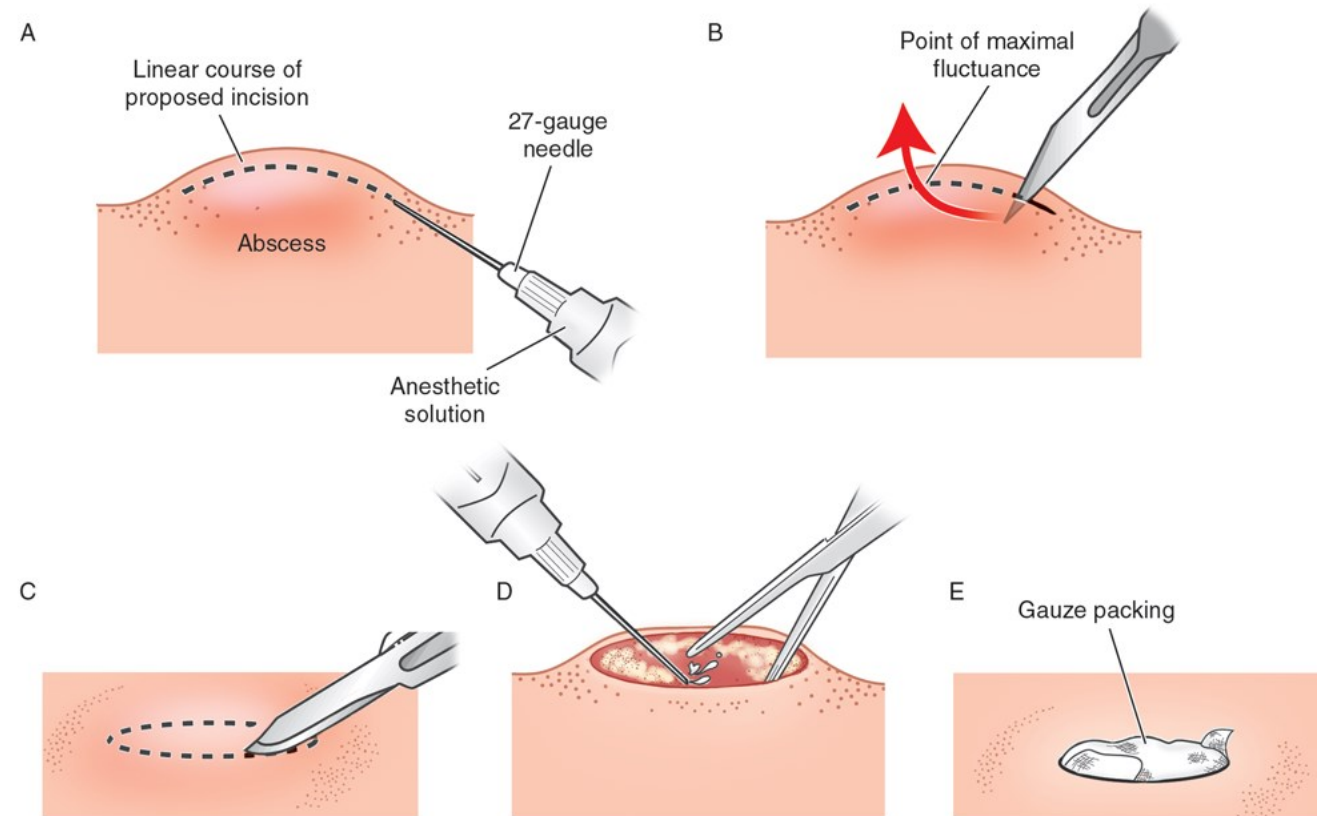
^aEMB, eosin–methylene blue agar; MC, MacConkey agar; SS, *Salmonella–Shigella* agar; BS, bismuth sulfite agar; HE, Hektoen enteric agar.

Source: Adapted from Murray, P.R., E.J. Baron, J.H. Jorgenson, M.L. Landry, and M.A. Tenover. 2007. *Manual of Clinical Microbiology*, 8th edition. American Society for Microbiology, Washington, DC.

4. Wounds, Abscesses and Pus

❑ for the detection of pathogens associated with infections linked to traumatic injuries :

- Cuts
- Burns
- Bites
- penetration of foreign objects, etc.



Source: Reichman EF: *Emergency Medicine Procedures*, Second Edition: www.accessemergencymedicine.com
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4. Wounds, Abscesses and Pus

- ❑ very careful sampling must be done to exclude the normal flora
- ❑ after skin disinfection → aspirates from wound abscesses obtained

from the center and from the wall
- from wounds : at the base; with a curette

4. Wounds, Abscesses and Pus

❑ **curette** = a small, scoop-shaped surgical instrument used to scrape the walls of a body cavity



4. Wounds, Abscesses and Pus

- **since some wound pathogens are anaerobic → transport to the laboratory should be under anaerobic conditions**

4. Wounds, Abscesses and Pus

❑ common wound pathogens :

S. aureus (associated with purulent discharges)

Enteric bacteria

P. aeruginosa

Bacteroides

***Clostridium*, etc.**

Collecting Specimen for Wound Culture

<https://www.youtube.com/watch?v=A42aaHz9wm8>

5. Cerebrospinal Fluid (CSF)

- ❑ obtained in cases when meningitis or an infection of the CNS is strongly suspected



prompt and precise diagnosis is essential



**meningitis ranks high among the
medical emergencies → WHY???**

sequelae = abnormal conditions from disease
recovered (worse disease occurred after)

5. Cerebrospinal Fluid

- ❑ samples must be obtained carefully and processed immediately



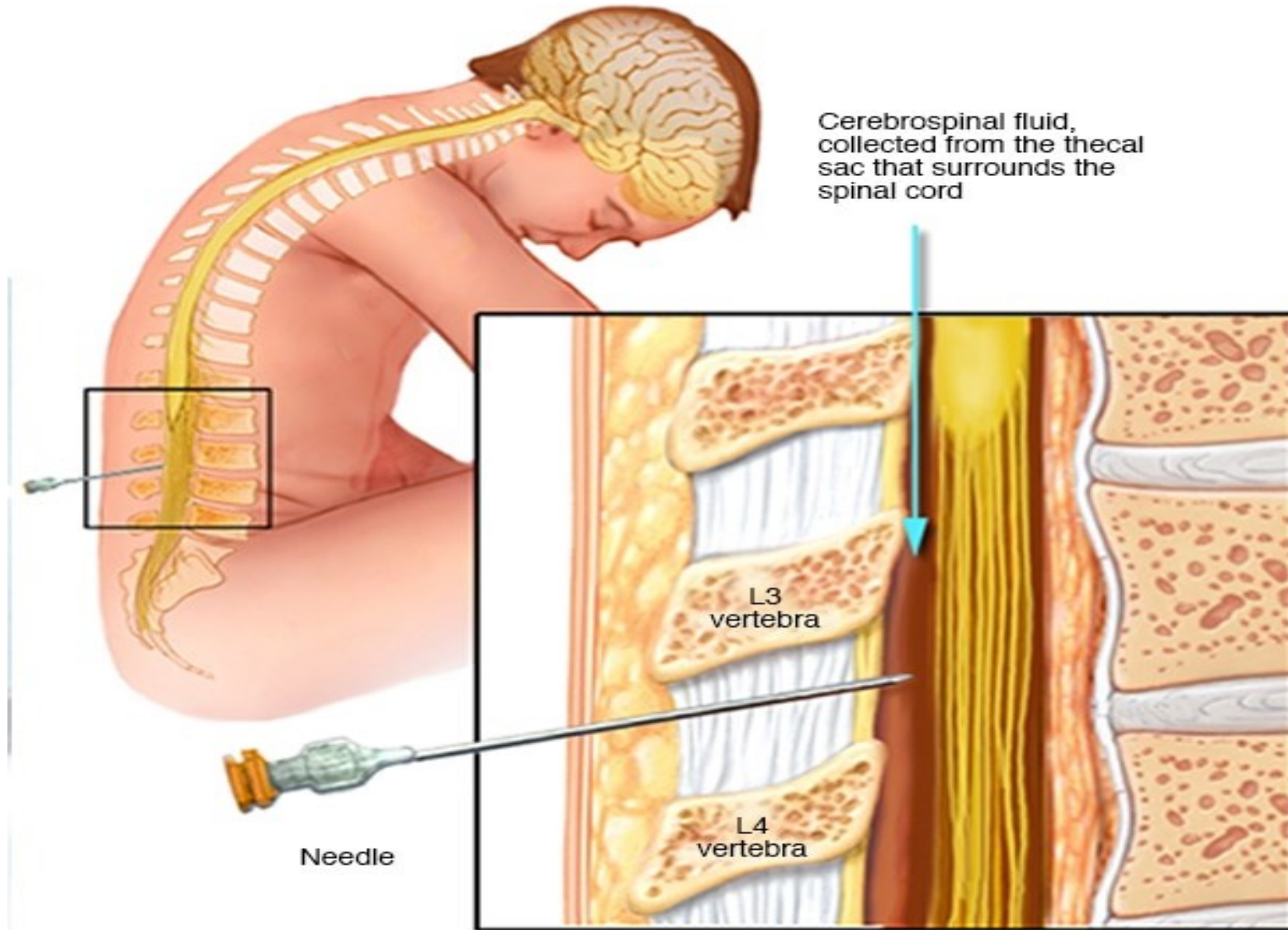
(never frozen or refrigerated because the pathogens are labile)



Neisseria meningitidis
Streptococcus pneumoniae
Haemophilus influenzae

- other microorganisms can also cause meningitis as long as it is already infected

5. Cerebrospinal Fluid



Issue
#1

LUMBAR PUNCTURE



spinal-tap method

5. Cerebrospinal Fluid

- ☐ for bacteria/viruses : 1-2 ml sample
for mycobacteria/fungi : > 2 ml
- ☐ ~ 3- 4 portions of 2-5 ml @ may be collected

6. Respiratory Secretions

□URT : throat specimens
nasopharyngeal specimens
middle ear aspirates, etc.

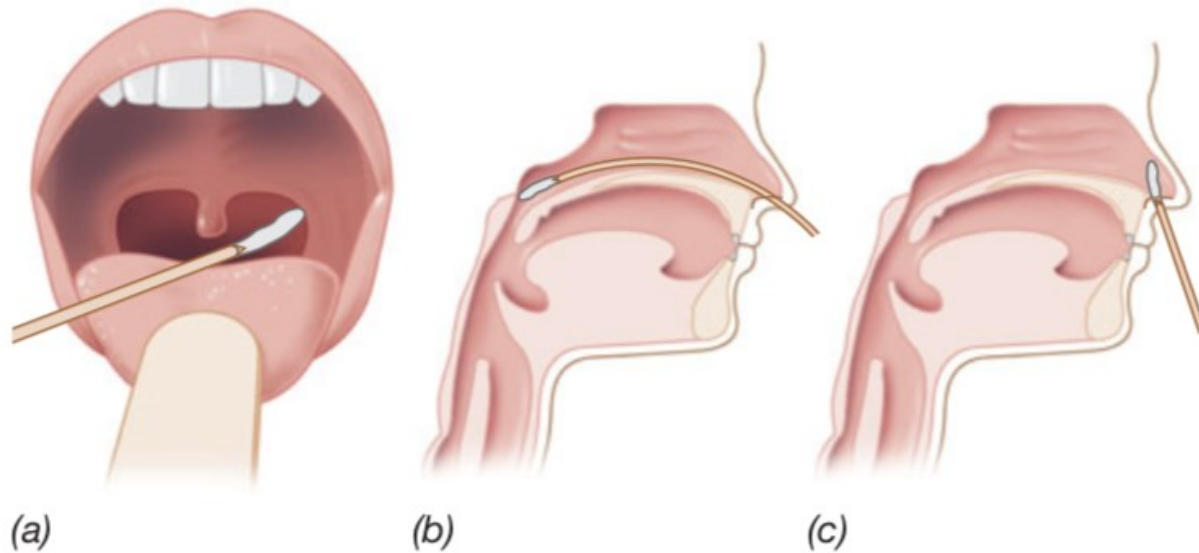


Figure 31.2 Methods for obtaining specimens from the upper respiratory tract. (a) Throat swab. (b) Nasopharyngeal swab passed through the nose. (c) Swabbing the inside of the nose.

□LRT : sputum
pleural fluid
lung tissue, etc.

Mercy rinses her mouth before the sputum sample is collected

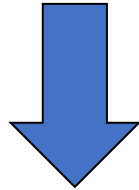


pleural fluid = water in the lungs

6. Respiratory Secretions

❑ For throat swab specimens :

→ contact with saliva must be avoided



can inhibit the recovery of Grp. A
streptococci (*S. pyogenes*) **or**
the pathogen may be overgrown by the normal flora

6. Respiratory Secretions

❑ For throat swab specimens :

Group A streptococci → the most important agents of bacterial pharyngitis or “strep throat”

However, most sore throats are due more to viral infections (~ 90-95% in adults; 80-85% in children)

6. Respiratory Secretions

□throat swab specimens

→ collected in suspected cases of :

pneumonia

is not only caused by bacteria but also inflamed alveoli = filled with pus or liquid = difficulty in breathing

diphtheria

is caused by *corynebacterium diphtheriae*

pertussis

aka "whooping cough"

**sinusitis (by *M. pneumoniae*,
H. influenzae, *S. aureus*, etc.)**

- inflammation of sinuses

RT infections by viruses

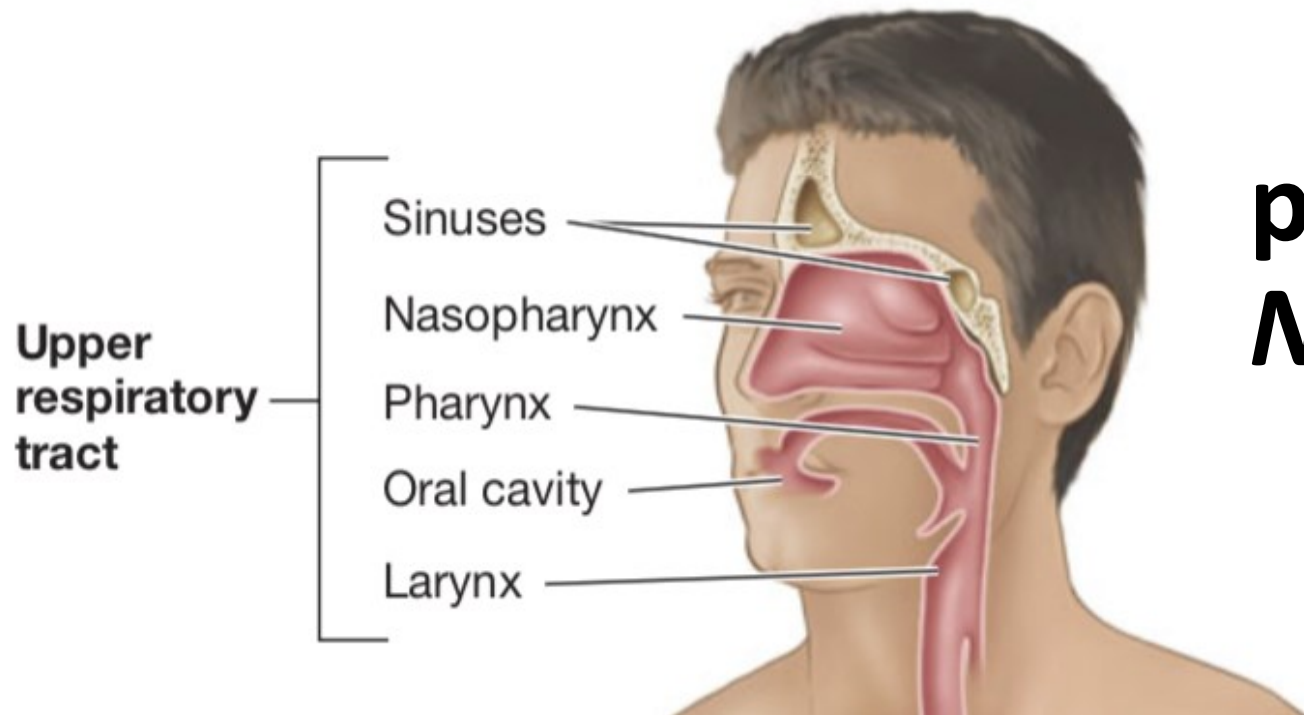
6. Respiratory Secretions

opening wide while swabbing will result to more accessible sphincter-like in the throat

❑ Nasopharyngeal specimens

may be cultured → in suspected cases of :

pertussis
***N. meningitidis* carriage**



6. Respiratory Secretions

❑ Middle ear aspirates (myringotomy)

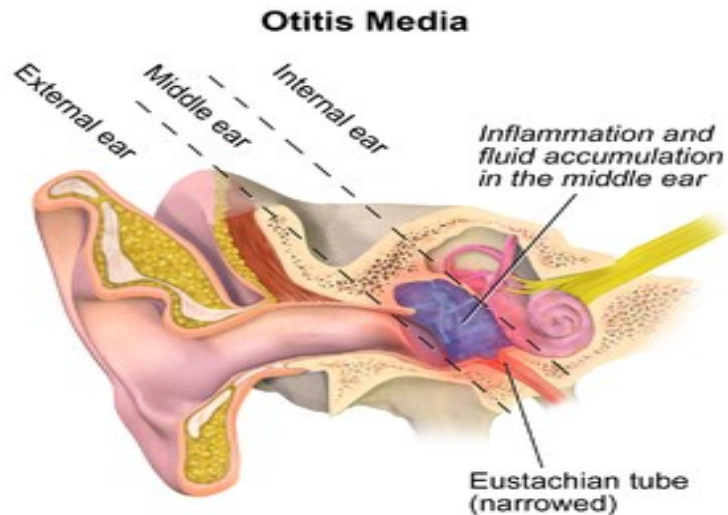
→ in cases of otitis media

- difficulty in balancing while walking if there's too much fluid in the ears

Otitis media = inflammation of the middle ear

= usually due to bacterial/viral infection

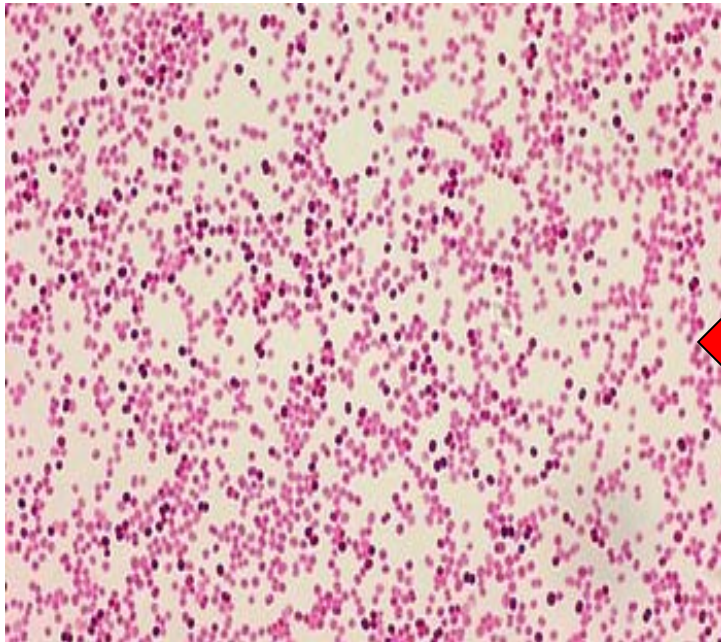
= may be with fluid accumulation



6. Respiratory Secretions

❑ Middle ear aspirates

→ In about 30-50% of aspirated fluids in otitis media, the implicated pathogens :



pneumococci

H. influenzae

Moraxella catarrhalis

- aside from otitis media, it can also infect eyes, central nervous system and joints in humans

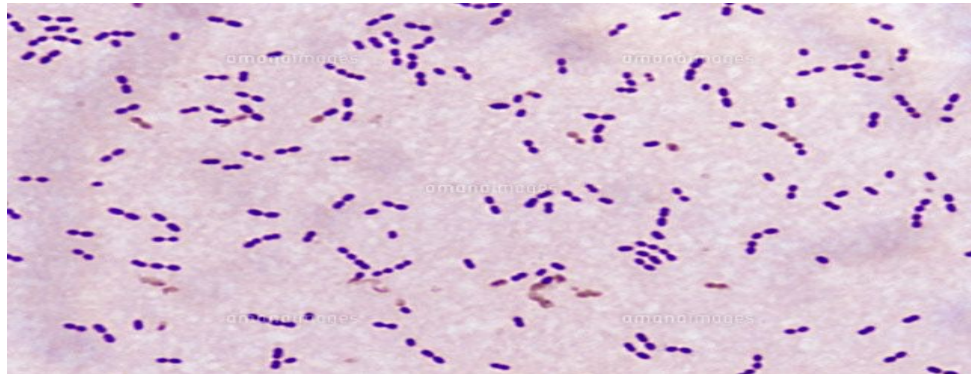
hemolytic streptococci

6. Respiratory Secretions

❑ For lower RT specimens

→ care must be taken to avoid contamination with saliva and mouth flora (rinse mouth first; only specimen from a deep cough)

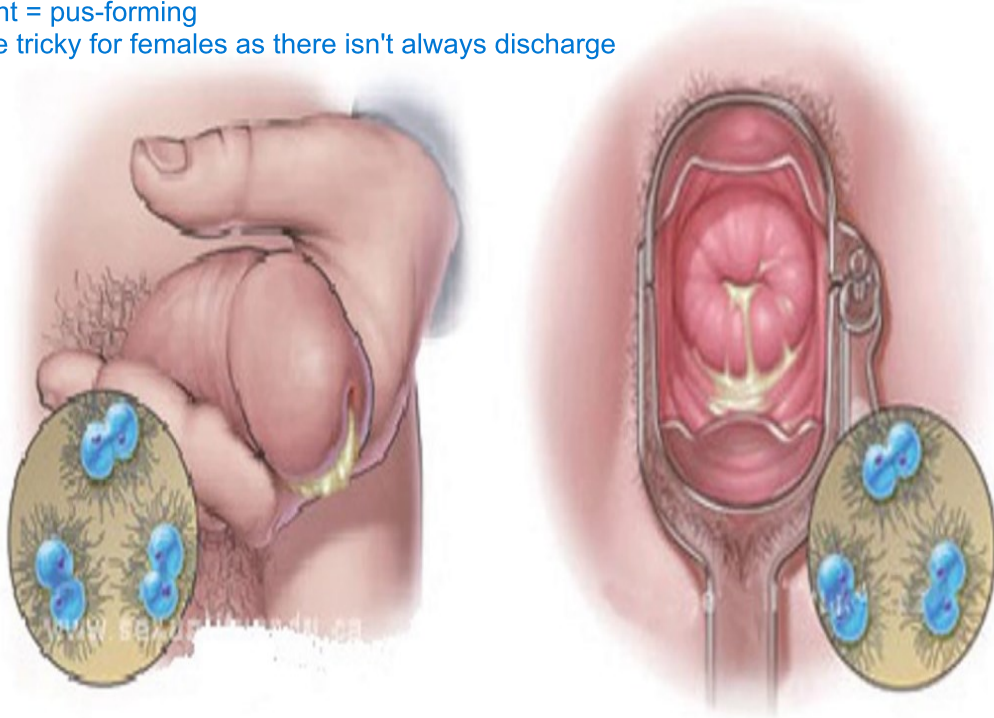
→ pneumococci : the most frequent cause of community-acquired bacterial infections



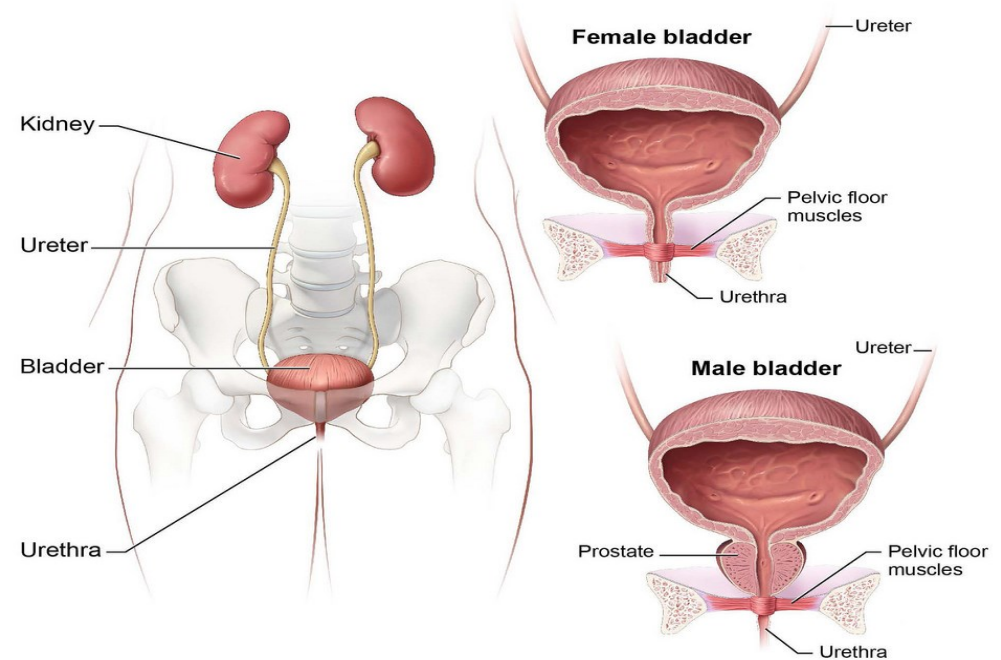
7. Genital Specimens

□ for the detection of pathogens causing Sexually Transmitted Infections (STIs)

- purulent = pus-forming
- may be tricky for females as there isn't always discharge



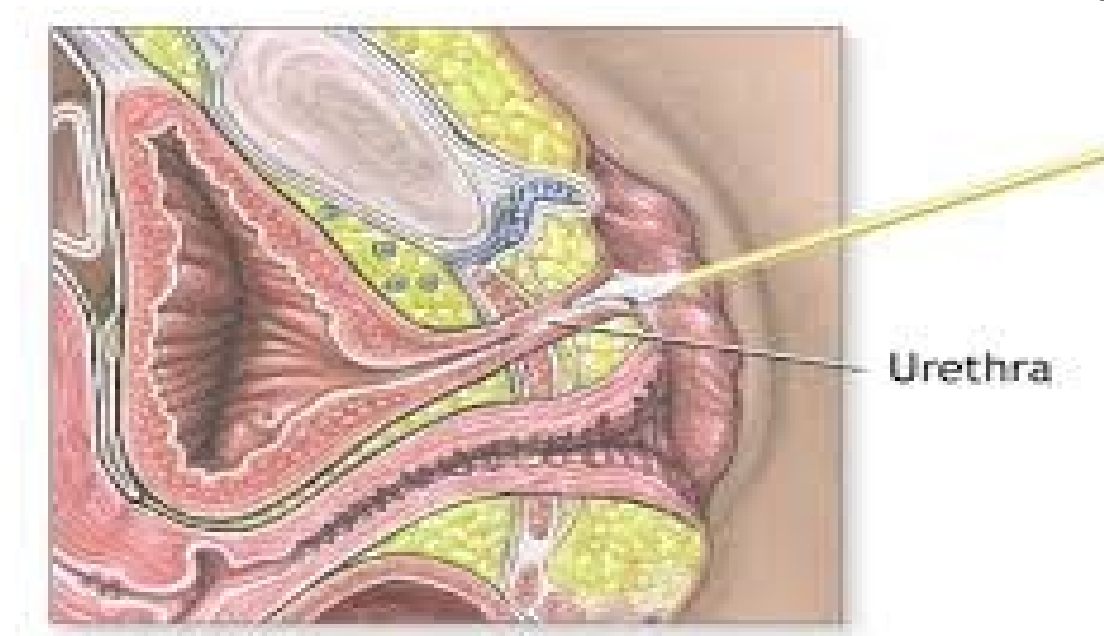
- In males, a purulent urethral discharge is usually indicative of STI.



- In females, samples are usually obtained by swab from the cervix and the urethra

7. Genital Specimens

- include :
 - urethral/cervical exudates
 - vaginal discharges
 - swabs of inflamed area, etc.



Urethral discharge is collected
on a cotton swab

Wet Mount, KOH and Cervical Sampling

yeast forms - presence hyphae
trichomonas - motile flagella

7. Genital Specimens

□ STIs are classified as nongonococcal or gonococcal urethritis.

- Nongonococcal urethritis is usually caused by *C. trachomatis*, *Ureaplasma urealyticum*, or *Trichomonas vaginalis*.

- other STIs are part of nongonococcal

- Gonococcal urethritis is caused by *N. gonorrhoeae*.

- the only one microorganism responsible for gonococcal

General Techniques/Procedures

- Microscopy
- Culture isolation and pathogen ID
- Detection of antigen(s) from the pathogen using immunologic assays - immunological assay = use of serum (clear part of the blood); microorganisms are identified as antigens, serums are antibodies
- Detection of genes specific for the pathogen in specimens obtained from the patient (molecular method)
- Detection of meaningful antibody- or cell-mediated immune responses to the agent

Clostridium septicum stained with Ab conjugated with fluorescein isothiocyanate

Direct immunofluorescence = fluorescently labeled antibody recognizes antigen, direct attachment; monoclonal antibodies are costly and difficult

Indirect immunofluorescence = fluorescently labeled antibody attaches to unlabeled antibody

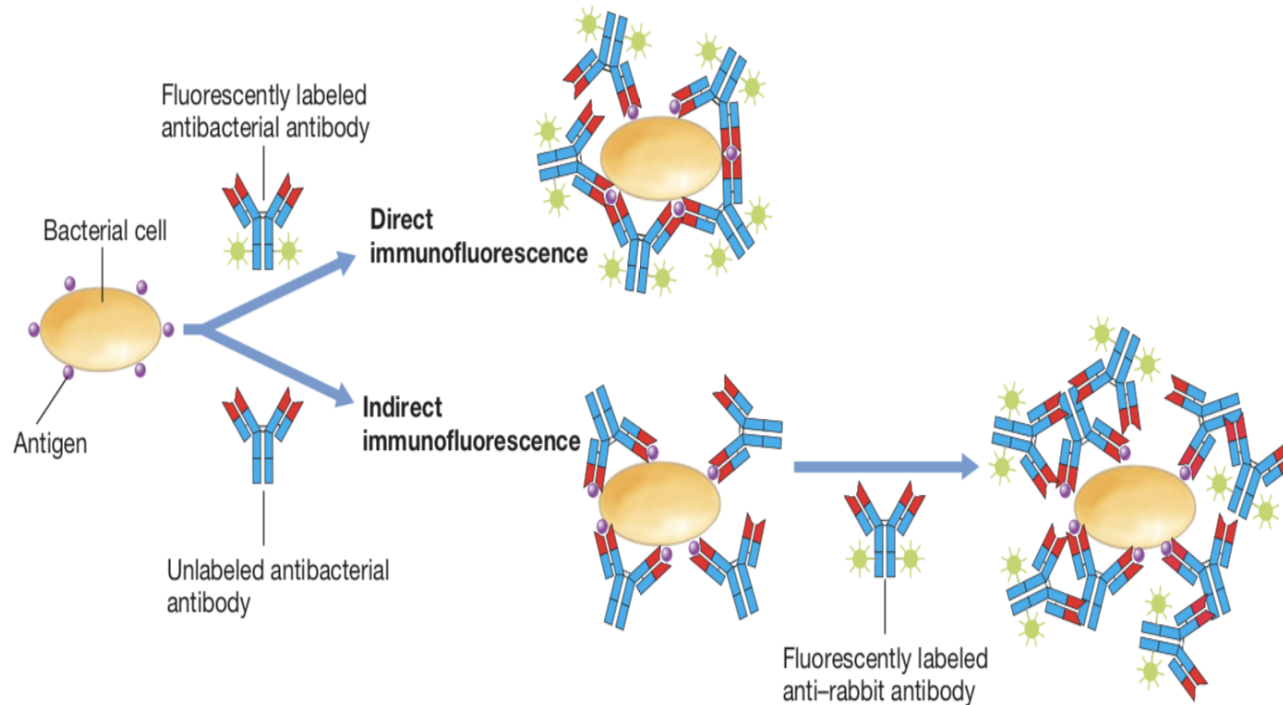


Figure 31.17 Fluorescent antibody methods for detection of microbial surface antigens.

Specificity - specific antibody recognize specific antigen



Wellcome Research Laboratories

Clostridium chauvoei stained with Ab conjugated with rhodamine B

- rhodamine B = usually entails red color

Fluorescent Antibody Reactions - reaction between an organism and antibody

Enzyme-Linked Immunosorbent Assay (ELISA) to Measure Specific Serum Antibodies

**animated and narrated by
Cary Engleberg, M.D., Professor,
University of Michigan Medical School**

Procedure

1. Antibodies (Y) to virus bound to wells of microtiter plate

2. Add patient sample (secretions, serum, and so on) suspected of containing virus particles or virus antigens (●) and wash wells with buffer

3. Add antiviral antibody containing conjugated enzyme (E-Y)



4. Wash with buffer

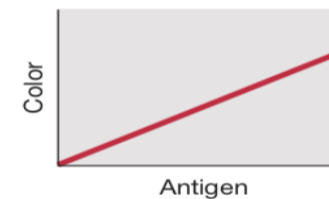
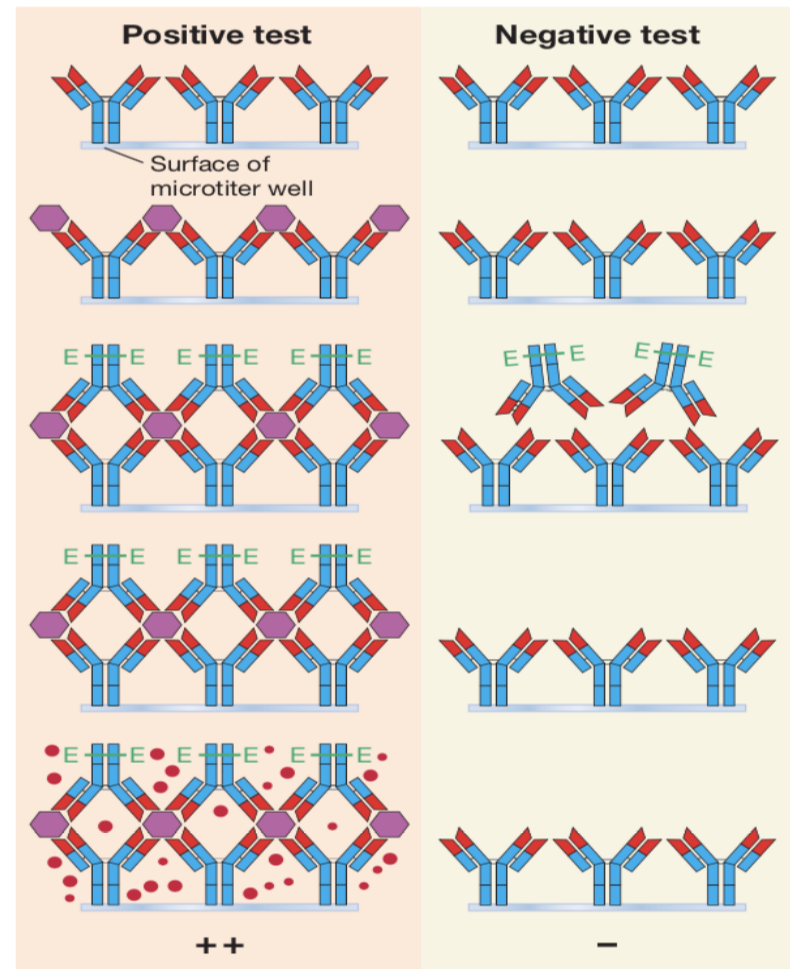
5. Add substrate for enzyme and measure amount of colored product (●)

Results

Colored product

Quantitation

Amount of colored product produced is proportional to amount of antigen



Immunologic Assay : e.g. ELISA (Enzyme-Linked Immunosorbent Assay)/ EIA (Enzyme immunoassay)

Microscopic Agglutination Test (MAT)

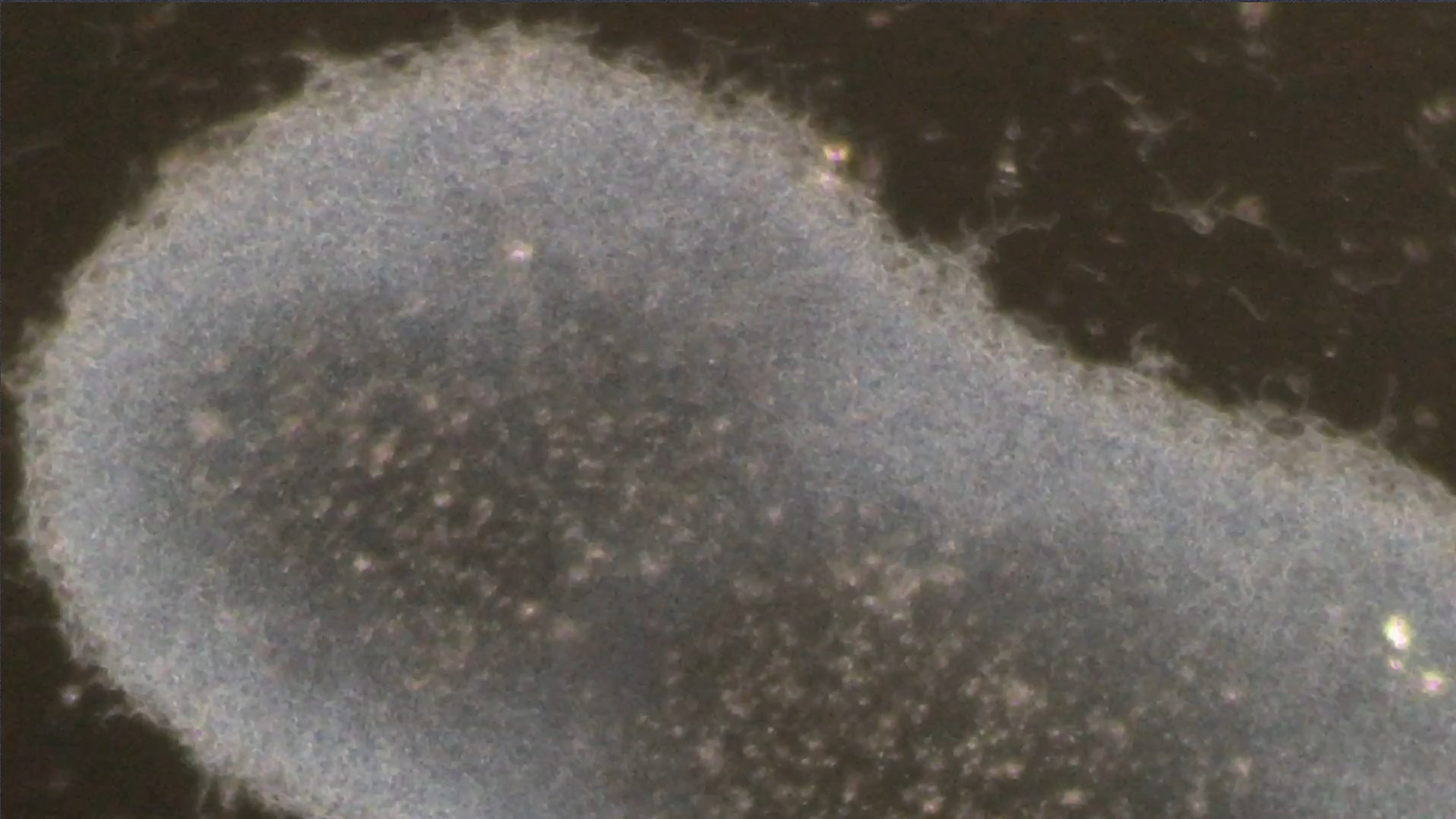
- looking under dark field microscope

- A test that determines agglutinating antibodies in the serum of a patient by mixing it in different dilutions with live or killed, formalized microorganism (e.g. *Leptospira*).

- agglutination = aclumping under dark field microscope

- Antibacterial antibodies present in the serum cause the bacteria to form clumps, a process called agglutination.
- Agglutination can be observed using a dark-field microscope. Agglutinating classes of antibodies can be both IgM and IgG

- IgM and IgG are the first antibodies



Leptospira spp.

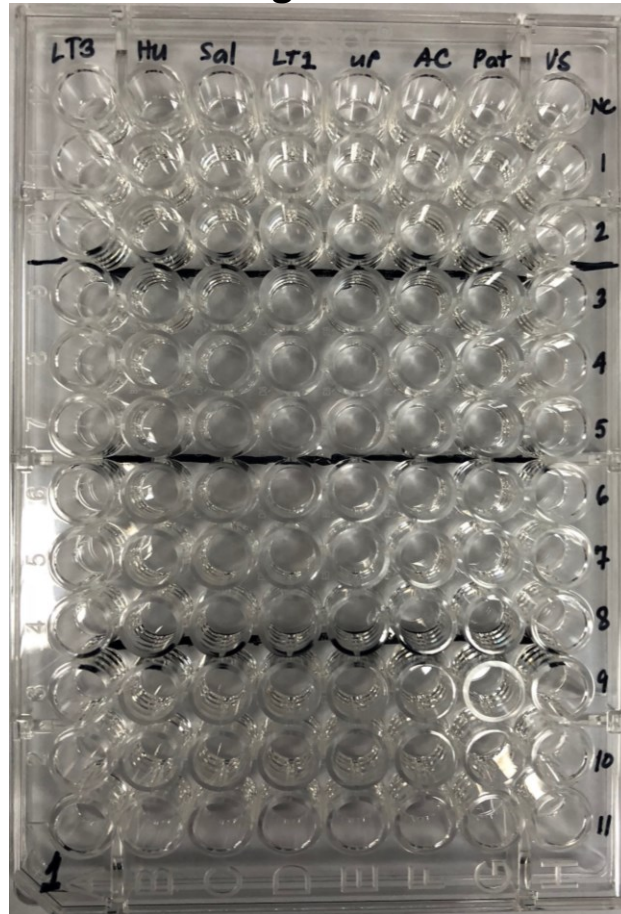
They clump due to antibodies present (agglutination)

Microscopic agglutination test (MAT) for humans

SCREENING MAT (Importance in epidemiology: to know which serovar of mcg infects people)

This is a 96 well microtiter plate

Antigen Panel



These are codes of serovars of mcgs (ex. leptospira)

vertical numbers = different patients

50 μ l dil. serum : 50 μ l Ag

Each serum was diluted to 1/25 in PBS

Human Sera Samples

- Cover plates and incubate at 30°C for 2-4 hours
- After incubation, prepare 10 μ l aliquot wet mount from each well
- view under a DFM at 400X mag.
- Aliquots that showed an agglutination of $\geq 50\%$ of leptospire or $\leq 50\%$ free leptospire are considered positive
- Ags that exhibited a positive reaction with the serum were further tested with quantitative MAT

Genomospecies	Serogroup	Serovar	Strain
<i>L. borgpetersenii</i>	Javanica	Javanica	K6*
		Javanica	Veldrat Batavia 46
		Poi	Poi
	Tarassovi	Tarassovi	Perepelitsin
	Autumnalis	Ballum	Fort Bragg
	Celledoni	Anhoa	LT 90-68
<i>L. fainei</i>	Mini	Mini	Sari
	Sejroe	Sejroe	M84
<i>L. fainei</i>	Hurtsbridge	Hurtsbridge	BUT 6T
<i>L. noguchii</i>	Louisiana	Louisiana	LSU 1945
	Panama	Panama	CZ 214K
<i>L. alexanderi</i>	Manhao	Manhao 3	L60
<i>L. weilii</i>	Sarmin	Sarmin	Sarmin
<i>L. santarosai</i>	Shermani	Shermani	1342 K
<i>L. interrogans</i>	Pyrogenes	Manilae	LT 398*
		Manilae	K64*
		Pyrogenes	Salinem
	Canicola	Canicola	Hond Utrecht IV
	Autumnalis	Autumnalis	Akiyami A
	Bataviae	Losbanos	LT 101-69*
		Losbanos	K37*
	Hebdomadis	Hebdomadis	Akiyami B
	Australis	Australis	Akiyami C
		Copenhageni	M20
	Icterohaemorrhagiae	Icterohaemorrhagiae	RGA
		Icterohaemorrhagiae	Ictero No. 1
	Pomona	Pomona	Pomona
	Sejroe	Hardjo	Hardjoprajitno
<i>L. kirschneri</i>	Djasiman	Djasiman	Djasiman
	Grippotyphosa	Grippotyphosa	K5*
		Ratnapura	UP-BL-FR-13*
<i>L. kirschneri</i>	Grippotyphosa	Grippotyphosa	Moskva V
	Cynopteri	Cynopteri	3522 C
<i>L. meyeri</i>	Samaranga	Samaranga	Veldrat Semarang 173
<i>L. biflexa</i>	Samaranga	Patoc	Patoc 1
	Andamana	Andamana	CH11
<i>L. yanagawae</i>	Samaranga	Saopaulo	Sao paulo

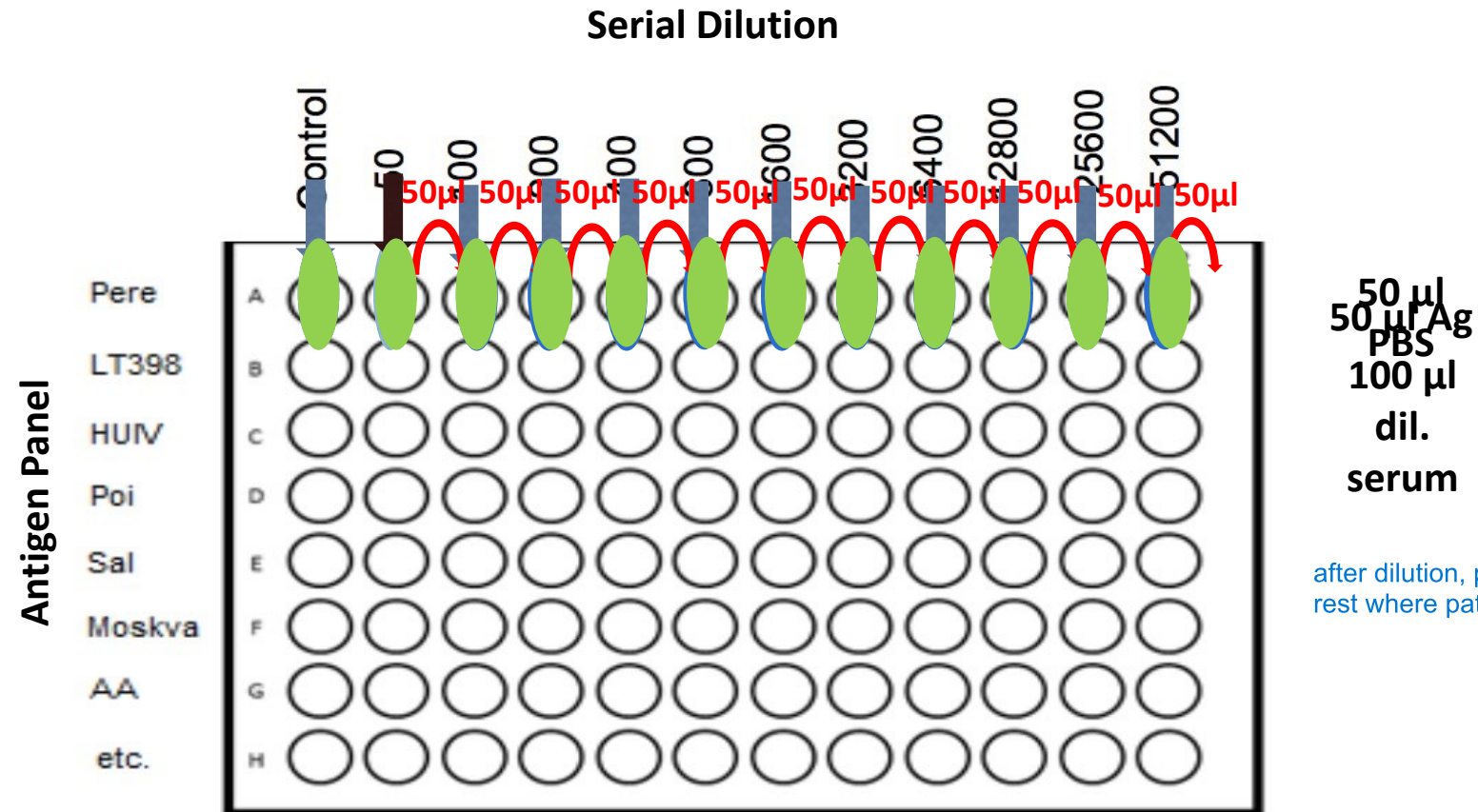
Legend: *Local isolates

Panel of *Leptospira* strains used as antigens in MAT

In epidemiology, we can identify in the serovar the source of infection
 rats can be carried via cargo, air, car, etc.
 not only rats can be carriers of leptospirosis; humans, cats and dogs too

Microscopic agglutination test (MAT) for humans (QUANTITATIVE MAT)

- reporting should be: which serovar produced the highest titer
- highest dilution = less amount of antibodies



after dilution, place antigen = then perform the rest where patient tested positive

Titer used: $\geq 1:100$

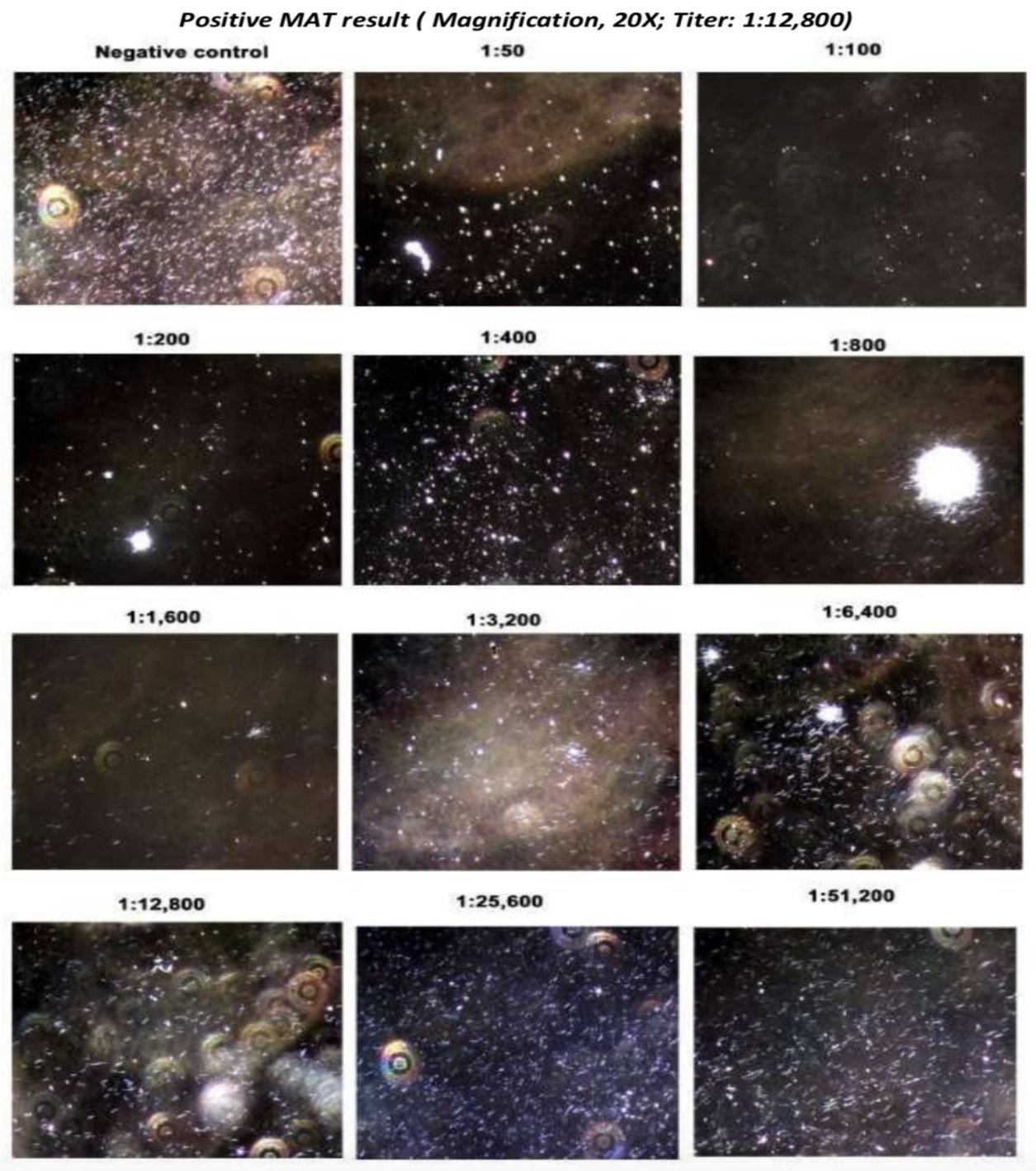
active leptospirosis = titer must not be lower than 1:400

Microscopic agglutination test (MAT) for humans

QUANTITATIVE MAT

- 
- Cover plates and incubate at 30°C for 2-4 hours
 - After incubation, prepare 10 µl aliquot wet mount from each well
 - View under a DFM at 400X mag.
 - Aliquots that showed an agglutination of $\geq 50\%$ of leptospire or $\leq 50\%$ free leptospire are considered positive
 - Seropositive samples are those whose aliquot tested positive with $\geq 1:400$ titer.

free leptospirosis = no antibodies, no serum



General Techniques/Procedures

- Microscopy
- Culture isolation and pathogen ID
- Detection of antigen(s)/antibodies using immunologic assays
- Detection of genes specific for the pathogen in specimens obtained from the patient (molecular method)
- Other Tests

A known nucleic acid probe (DNA or RNA) is hybridized to a specific nucleic acid sequence in a test sample and the paired hybrid is then detected (e.g. PCR assay)

Sometimes, there are micororganisms we cannot detect using cultures, serological assays, etc. Thus we have molecular methods which are faster

PCR detection of pathogens :

detection of birulent genes

Staphylococcus aureus

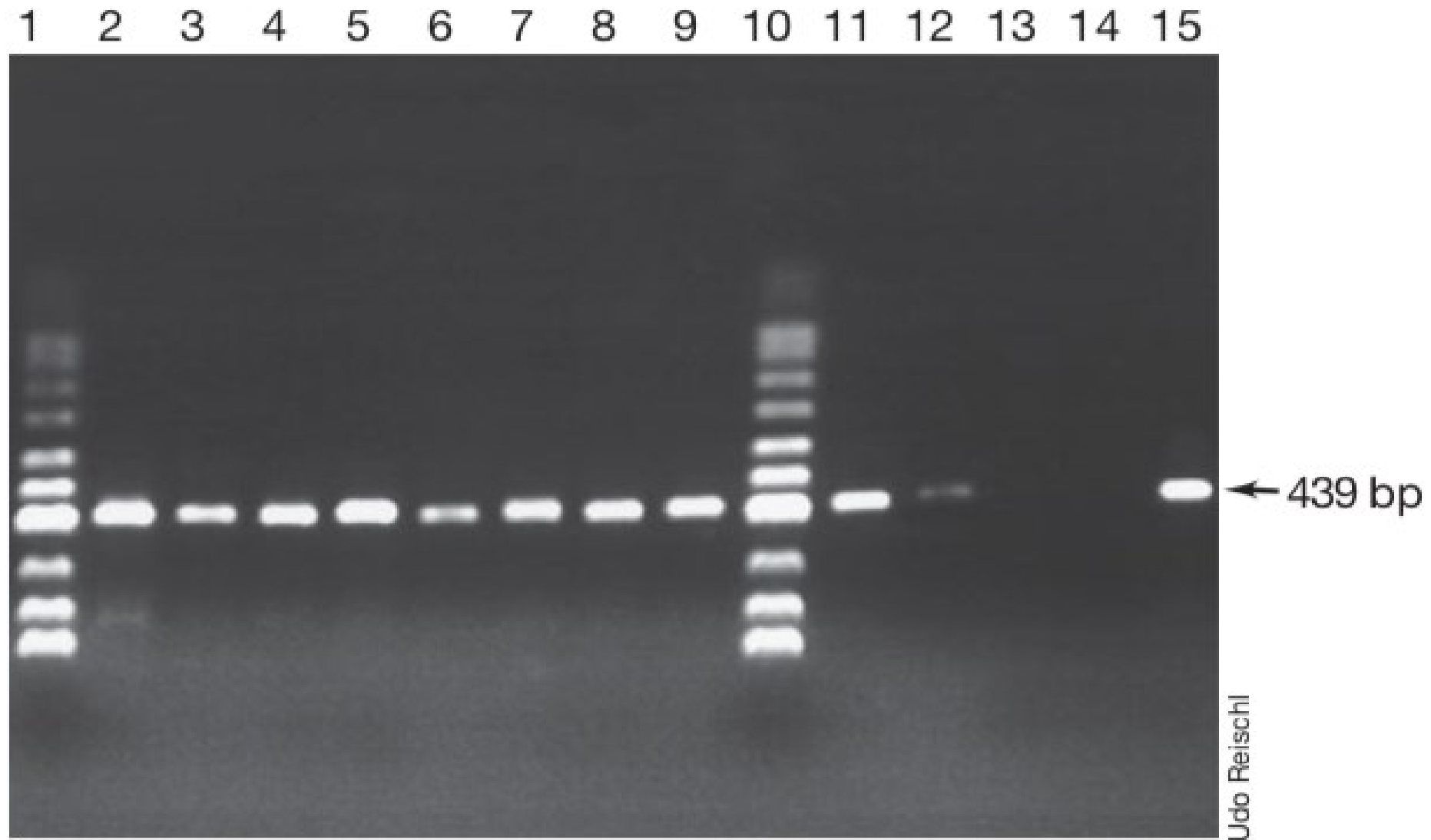
Streptococcus pneumoniae is hard to culture; also publicly acquired

Neisseria gonorrhoeae is publicly acquired

Clostridium perfringens, difficult to culture because it is anaerobic

***Mycobacterium tuberculosis* etc.**

PCR analysis of patient sputum for *M. tuberculosis* in the diagnosis of tuberculosis



faint band = exposure only

General Techniques/Procedures

- Microscopy
- Culture isolation and pathogen ID
- Detection of antigen(s)/antibodies using immunologic assays
- Detection of genes specific for the pathogen in specimens obtained from the patient (molecular method)
- Other Tests

Other Tests

- Antimicrobial Susceptibility Testing:
organism is tested vs. a battery
of antimicrobials

MIC Determination (Minimum inhibitory concentration) - inhibit only

MBC (or MFC) Determinations, etc.

(Minimum bactericidal concentration and Minimum fungicidal concentration = 99.9% will be killed) - antibiotics are diluted

- Schlichter Test
 - type of antimicrobial susceptibility test for determining the titer of serum that has effective bactericidal activity
 - serum is diluted



MIC is preferred for cost-effectivity
MIC answers which lowest concentration can kill?
Turbidity is a sign of growth
lowest concentration = first clearing

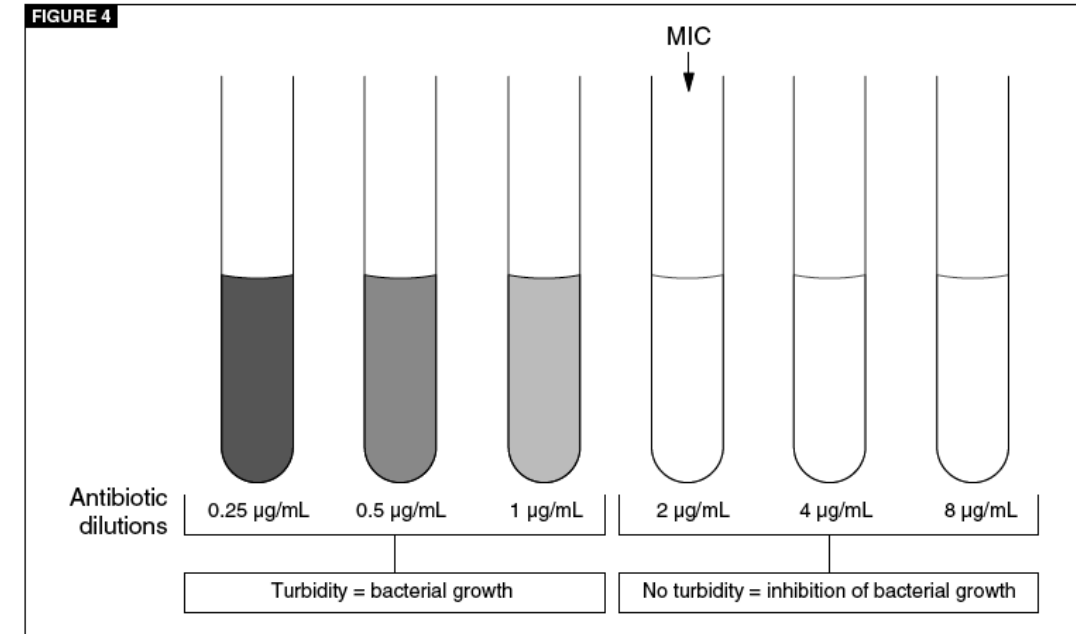
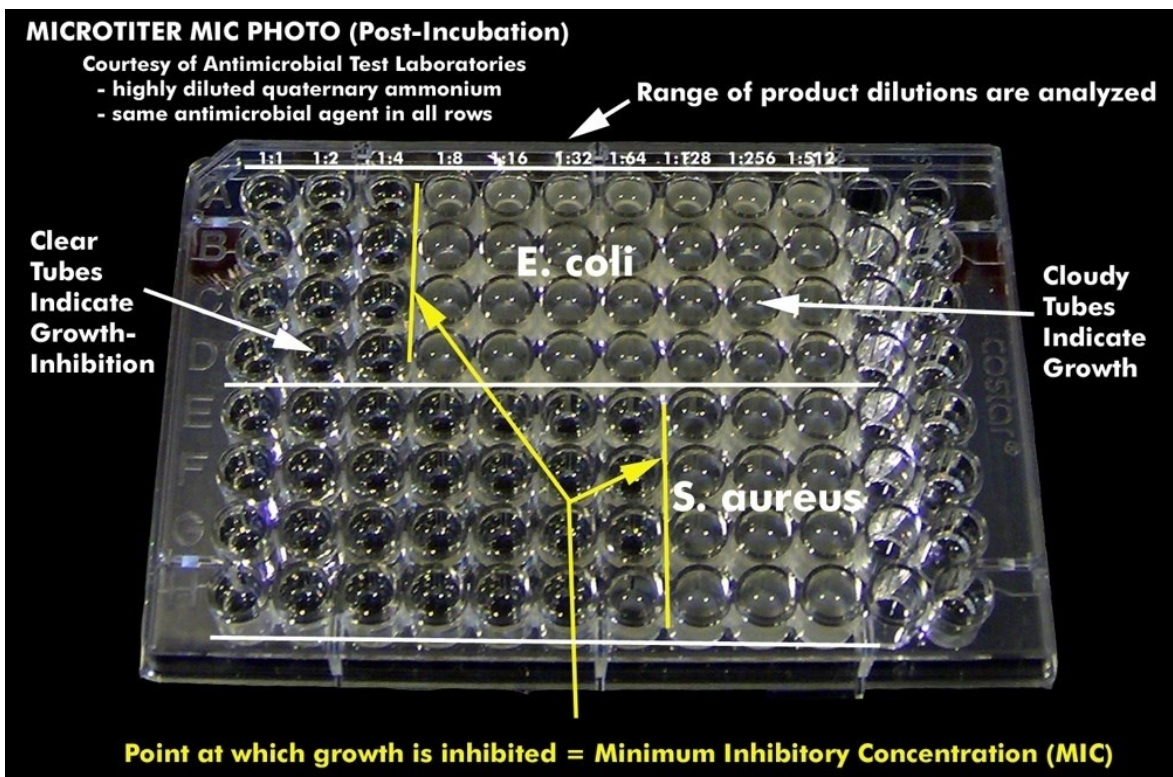


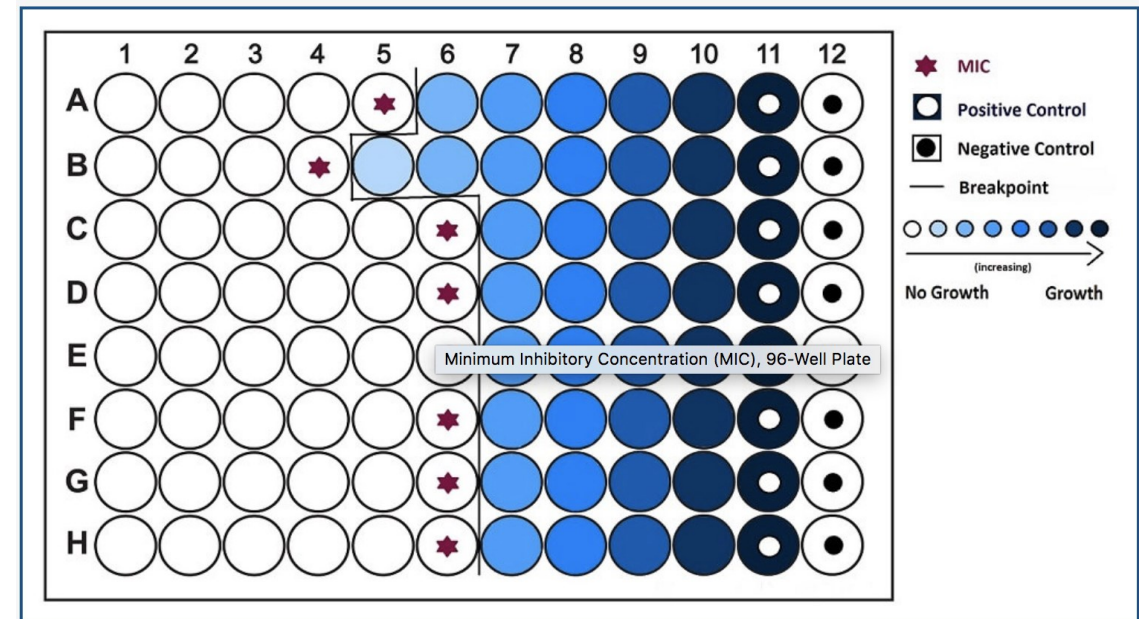
Diagram of the broth dilution method. The MIC is the lowest concentration of antibiotic that lacks turbidity (i.e., has no bacterial growth).

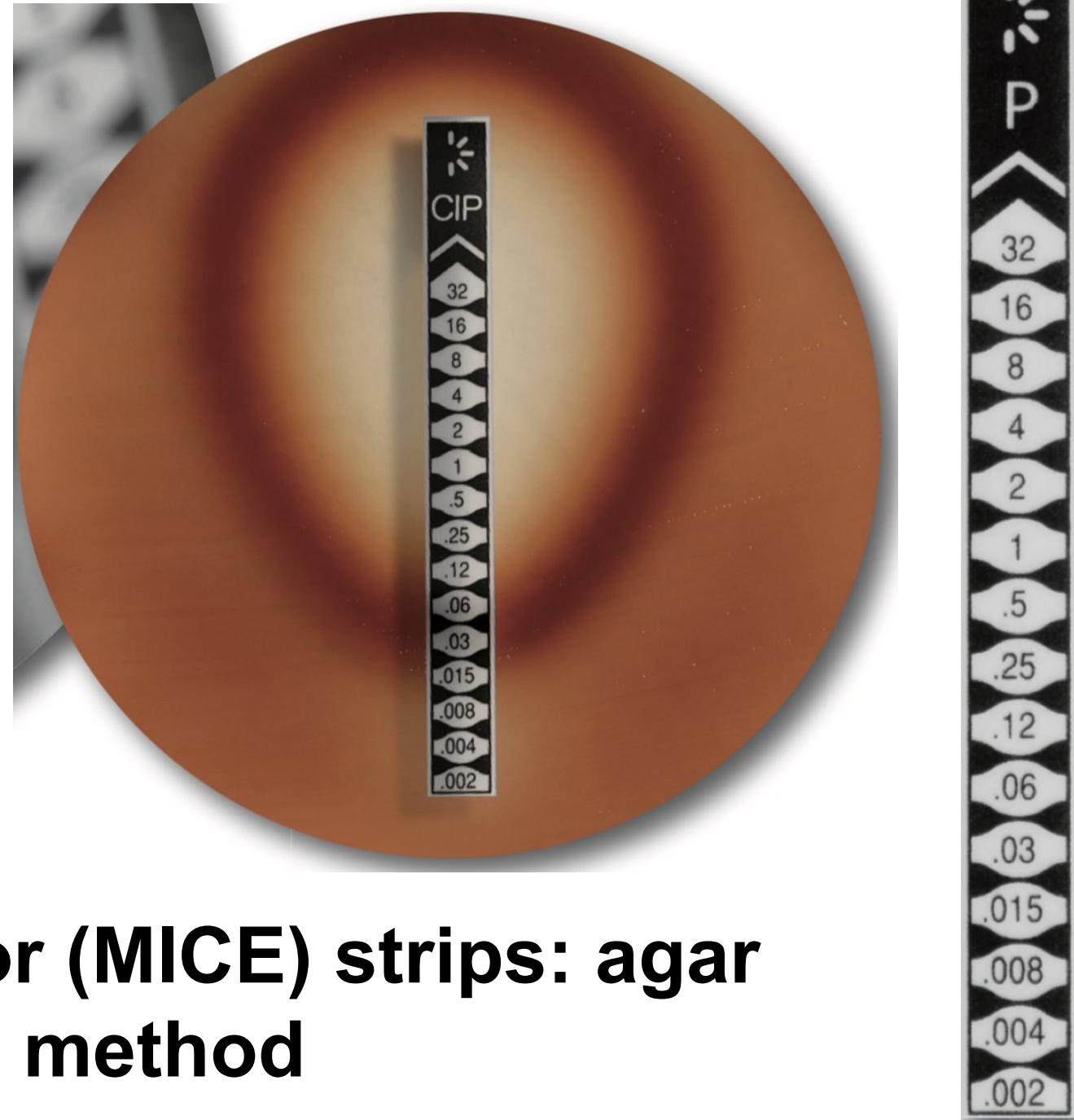
MIC Determination : broth dilution method (Antimicrobial susceptibility testing)



MBC/MFC is the lowest concentration of an antibacterial drug that results in a 99.9% reduction in the initial microbial density

This is an illustration of a panel representing Broth Microdilution MIC for 8 antimicrobials tested against a susceptible bacterial strain.

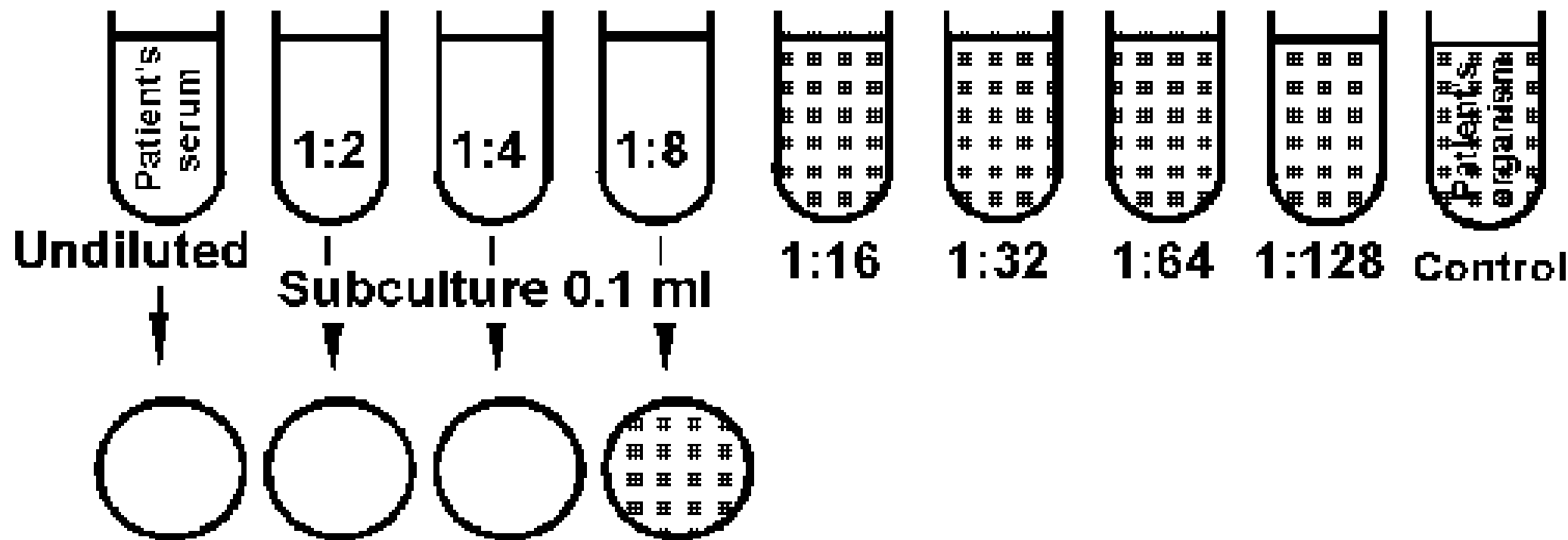




Oxoid's MIC Evaluator (MICE) strips: agar diffusion method

- MIC = first to produce zone of inhibition

Serial dilutions of patient's serum inoculated with patient's infecting organism



Schlichter Test : to determine what dilution of the serum is bactericidal to the infecting organism
: for patients receiving antibiotic therapy

BIOSAFETY

Objective: reduce or eliminate accidental exposure to or release of potentially hazardous agents

**CONTAINMENT = ...safe methods for
managing infectious
agents ... where these
are handled/maintained**

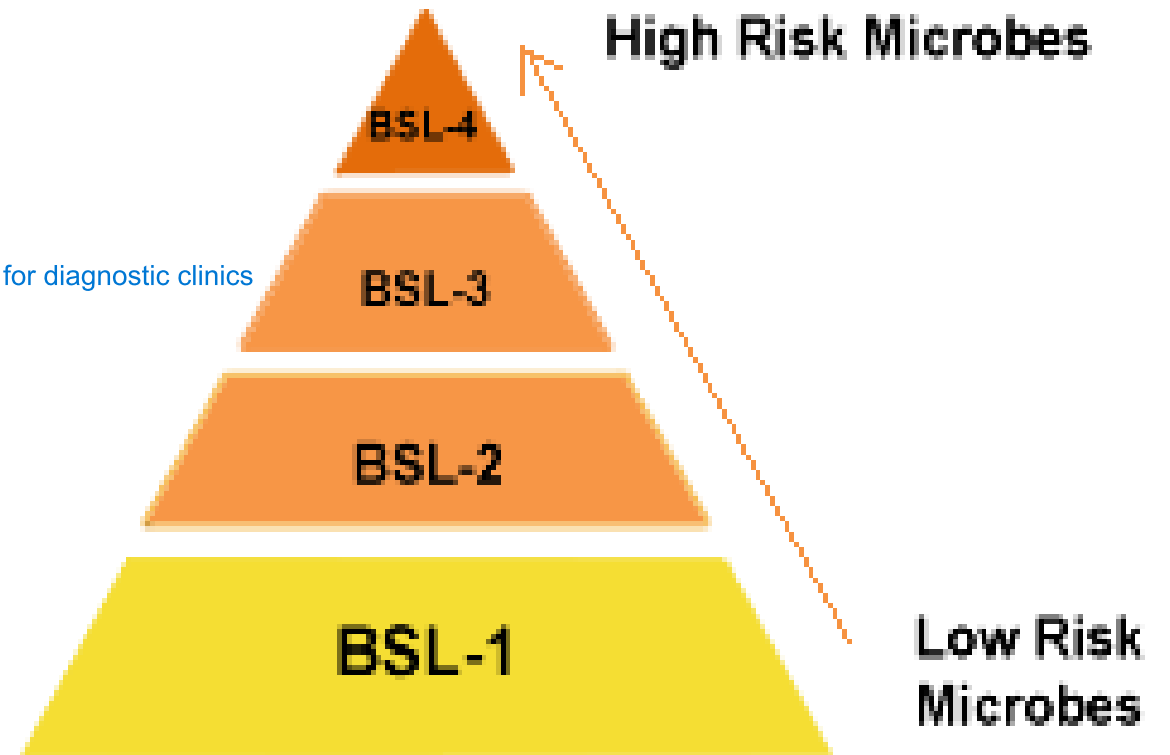
Safety Equipment (1° barriers) :

- **Biological safety cabinets (BSCs)** - although not required in university laboratories
- **Items for personal protection (PPE)**

BIOSAFETY

Facility Design (2° barriers) :

- ☐ Basic laboratory - BSL 1 AND BSL 2 are used for diagnostic clinics
- ☐ Containment laboratory
- ☐ Maximum containment laboratory



Biosafety Levels

• Biosafety Level 1

- Suitable for work involving well-characterized agents not known to cause disease in healthy adult humans and of minimal potential hazard to laboratory personnel and the environment.
- Examples:
 - *Bacillus subtilis*
 - *Naegleria gruberi*
 - Infectious canine hepatitis virus
 - *E. coli*

• Biosafety Level 2

- Suitable for work involving agents of moderate potential hazard to personnel and the environment
- Examples:
 - Measles virus
 - Salmonellae
 - Toxoplasma species
 - Hepatitis B virus

• Biosafety Level 3

- Suitable for work with infectious agents which may cause serious or potentially lethal disease as a result of exposure by the inhalation route.
- Examples:
 - *Mycobacterium tuberculosis*
 - St. Louis encephalitis virus
 - *Coxiella burnetii*

• Biosafety Level 4

- Suitable for work with dangerous and exotic agents that pose a high individual risk of aerosol transmitted laboratory infections and life-threatening disease.
- Examples:
 - Ebola Zaire virus
 - Rift Valley Fever virus

- *Coxiella burnetii* causes abortion (spoiled milk)
- *Mycobacterium tuberculosis* = aerosol
- St. Louis encephalitis = worst kind of dengue

Table 2. Relation of risk groups to biosafety levels, practices and equipment

RISK GROUP	BIOSAFETY LEVEL	LABORATORY TYPE	LABORATORY PRACTICES	SAFETY EQUIPMENT
1	Basic – Biosafety Level 1	Basic teaching, research	GMT GMT = Good microbiological techniques	None; open bench work
2	Basic – Biosafety Level 2	Primary health services; diagnostic services, research	GMT plus protective clothing, biohazard sign	Open bench plus BSC for potential aerosols
3	Containment – Biosafety Level 3	Special diagnostic services, research	As Level 2 plus special clothing, controlled access, directional airflow	BSC and/or other primary devices for all activities
4	Maximum containment – Biosafety Level 4	Dangerous pathogen units	As Level 3 plus airlock entry, shower exit, special waste disposal	Class III BSC, or positive pressure suits in conjunction with Class II BSCs, double-ended autoclave (through the wall), filtered air

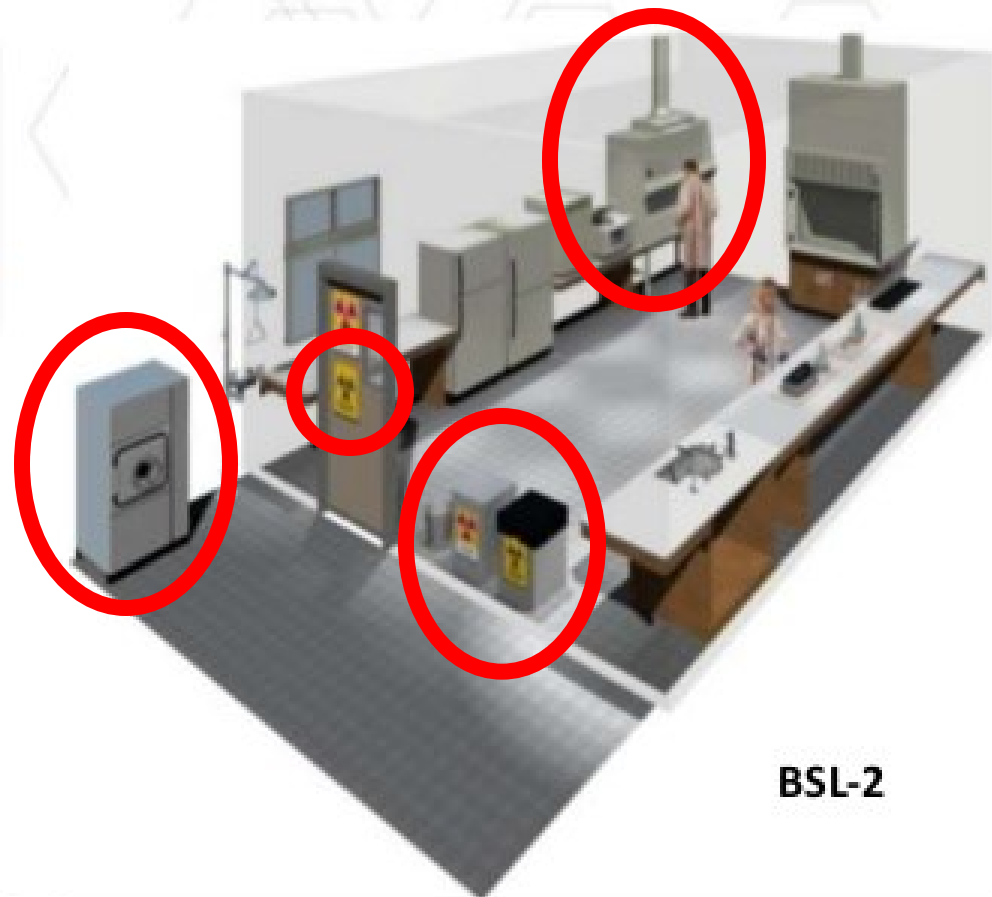
BSC, biological safety cabinet; GMT, good microbiological techniques (see Part IV of this manual)

From: WHO LBM 3rd edition

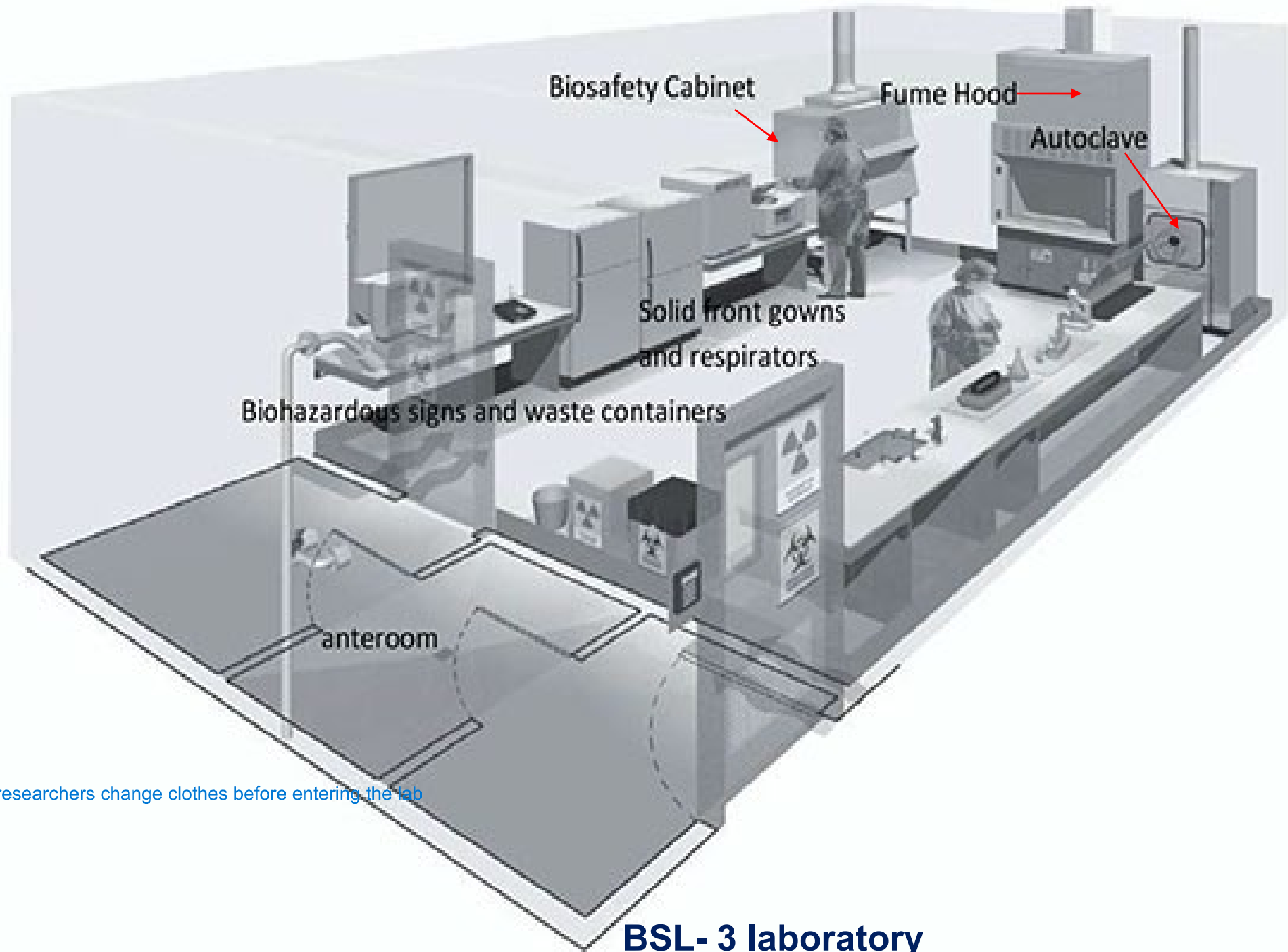
Biosafety Levels 1 & 2



BSL-1

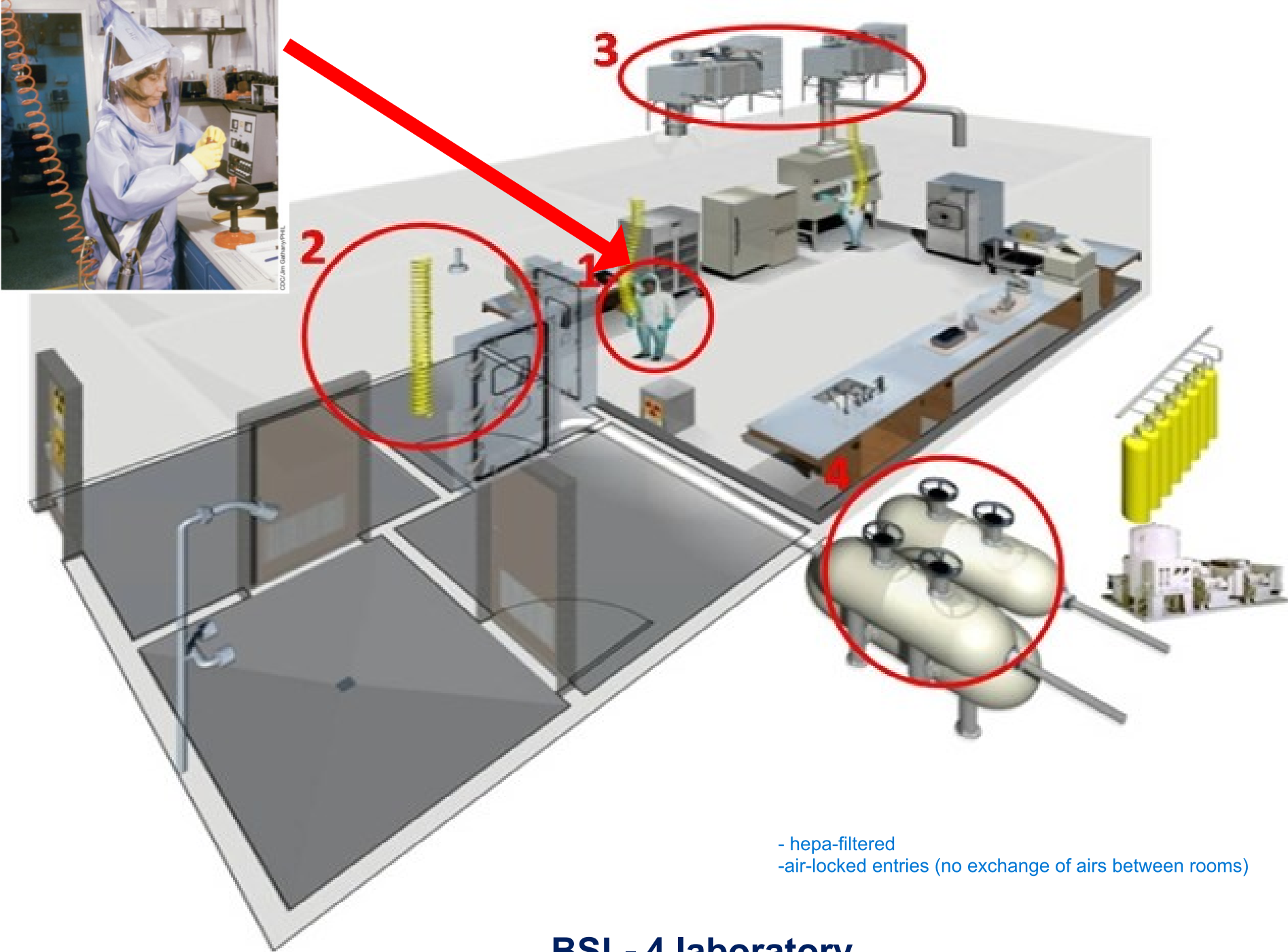


BSL-2



anteroom = where researchers change clothes before entering the lab

BSL- 3 laboratory



- hepa-filtered
- air-locked entries (no exchange of airs between rooms)

BSL- 4 laboratory

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